UNIT I

Introduction, Failures, LEFM / EPFM & Fatigue Crack Growth

EFM course outline | spectacular failures | LEFM vs EPFM | fatigue crack growth model

I. EFM Course Outline

Fracture mechanics is the study of how cracks / flaws in a structure grow and when they cause **fracture**. It gives a quantitative link between applied stress, flaw size and material toughness.

Unit	Main topics
I	Failures, LEFM & EPFM intro, fatigue crack growth model
II	Crack growth & fracture mechanisms, Griffith theory, energy release rate
III	Elasticity review, Westergaard solution (Mode I), K-G relation
IV	Multi-parameter stress field, mixed modes, SIFs, Irwin & Dugdale models
V	Toughness testing, Paris law, closure, J-integral, FAD, mixed mode, arrest & repair

Why study Fracture Mechanics ?

> Conventional design assumes a **flaw-free** material: design stress = σ_y / FoS. This ignores cracks.

> Real structures always contain flaws (welds, inclusions, machining marks, corrosion pits).

> Fracture can occur at stresses **far below yield** (e.g. Liberty ships, Comet).

> FM answers 3 questions: (1) residual strength with a crack, (2) critical crack size, (3) how long a crack takes to grow from a_0 to a_c (inspection interval).

I.1 Three variables of fracture design

Conventional approach has ONE variable (stress σ vs strength σ_y). The fracture mechanics approach has THREE:

Applied stress σ , Flaw size a , Fracture toughness K_c

- Fracture occurs when $K = \sigma Y \sqrt{(\pi a)} \geq K_c$ (K_c = material property, units MPa$\sqrt{m}$).
- Given any two variables, the third (critical) value can be found.
- **Fracture toughness K_c** = ability of a material containing a crack to resist fracture.

2. Spectacular Failures (Case Studies)

Learn these as: **what happened → cause → lesson**. Very common 5/10 mark question.

Failure / year	Cause (fracture mechanics view)	Lesson
Liberty ships (USA, 1943-47) ~2700 welded ships	Brittle fracture: welded hull (cracks cross plates), notch-brittle steel below DBTT in cold seas, stress concentration at square hatch corners. ~1500 serious fractures, about a dozen ships lost; some broke in two while docked.	Material toughness & DBTT; crack arresters; rounded hatch corners; Charpy test specified
de Havilland Comet I (1953-54)	Two aircraft broke up in flight. **Fatigue cracks** started at corners of square ADF windows (high Kt) under repeated cabin pressurisation; riveted-hole stress concentration.	Fatigue + stress concentration; full-scale fatigue testing; rounded windows; fail-safe design
Silver Bridge, Ohio (1967)	Eyebar chain failed: a tiny crack in one eyebar grew by **stress-corrosion / corrosion fatigue** to critical size. 46 deaths.	Non-redundant (single load path) structure; inspection
F-111 (USAF, 1969)	Wing failed at well below design load from a pre-existing forging flaw in the wing pivot fitting (D6AC steel).	Led to **damage tolerance** requirement (assume initial flaws)
Alexander Kielland rig (Norway, 1980)	Fatigue crack from a fillet weld of a hydrophone support in a brace; brace failed, platform capsized. 123 deaths.	Weld fatigue, poor inspection, no redundancy
Aloha Airlines 243 (1988)	Fuselage upper section tore off: **multiple-site damage** (MSD) of lap-joint rivet holes, corrosion & disbonding in an ageing aircraft.	Ageing aircraft programme; MSD, widespread fatigue damage
Eschede train (Germany, 1998)	Fatigue crack in a wheel tyre of a high-speed train; 101 deaths.	Inspection & fatigue crack growth analysis
Titanic (1912)	Brittle steel (high S, P) + cold water below DBTT; rivet failures (slag-rich iron rivets) contributed.	Ductile-brittle transition & material choice

Common causes seen in all failures

> Pre-existing flaw or stress concentrator (sharp notch, weld defect, rivet hole).
> Low-toughness material or low service temperature (below DBTT).
> Cyclic loading (fatigue), corrosive environment (SCC, hydrogen embrittlement).
> Lack of redundancy and of regular NDT inspection.

2.1 Evolution of design philosophy

Philosophy	Idea	Example
Safe-life	Component retired after N_f / scatter factor; no cracks allowed	Landing gear, helicopter rotors
Fail-safe	Structure has redundant paths; failure of one member is carried by others	Aircraft wing, multi-spar
Damage tolerant	Assume an initial flaw a_0; predict growth $a_0 \rightarrow a_c$; inspect before a_c is reached	Modern aircraft, pressure vessels, pipelines

3. Introduction to Fracture Mechanics

3.1 Why stress concentration is not enough

- For an elliptical hole, $\sigma_{max} = \sigma(1 + 2a/b)$. For a sharp crack, $b \rightarrow 0$ so $\sigma_{max} \rightarrow \infty$ (stress singularity).
- A stress-based criterion ($\sigma_{max} = \sigma_u$) would predict failure at ~zero load – not true.
- Hence we need a **single-parameter** that characterises the crack-tip field: **K** (stress intensity factor) or **G** (energy release rate). Both are discussed in Units 2-3.

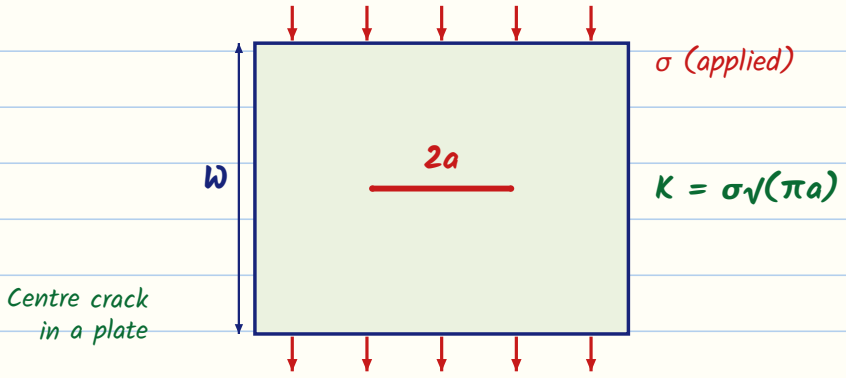


Fig 1.1 Plate with a central through crack under remote tension σ

3.2 Modes of crack loading

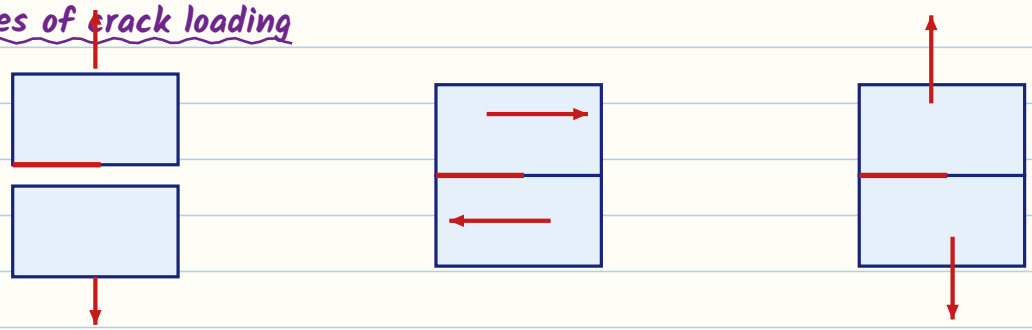


Fig 1.2 Three fundamental modes of fracture

- **Mode I** (opening): tensile stress ⊥ crack plane. Most important & most dangerous; K_I,

K_{Ic}.

- **Mode II** (sliding): in-plane shear, ⊥ to crack front. K_{II}.
- **Mode III** (tearing): out-of-plane shear, ∥ to crack front. K_{III}.
- Real cracks are often **mixed mode** (I + II + III) – superposition is applicable for linear elastic bodies.

3.3 Brittle vs ductile fracture

Brittle fracture	Ductile fracture
Little/no plastic deformation, sudden	Large plastic deformation, necking
Flat, shiny (cleavage) surface; chevron / river marks	Rough, dull, fibrous surface; cup-and-cone
Fast crack (~ 0.4 of sound speed), no warning	Slow, stable crack growth; warning given
Low energy absorbed; high strain-rate, low T, thick sections, triaxial stress	High energy absorbed; high T, thin sections

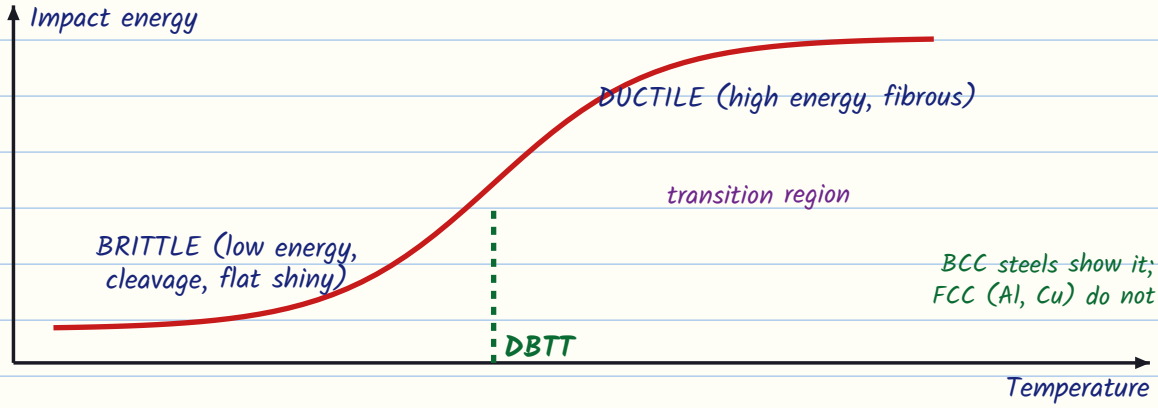


Fig 1.3 Ductile-to-brittle transition temperature (DBTT)

DBTT is found from the **Charpy V-notch impact** test (energy absorbed vs temperature). Steels in structures should operate above DBTT (e.g. ~ 27 J level).

4. LEFM and EPFM

4.1 LEFM – Linear Elastic Fracture Mechanics

- Applies when plastic zone at crack tip is **small** compared with crack length & specimen dimensions (small-scale yielding, SSY): $r_p << a, (W-a), B$.
- Whole body behaves linearly elastic; the crack-tip field is governed by **K** (or G).
- Assumptions: isotropic, homogeneous, linear elastic material; sharp crack; quasi-static loading; stress singularity ∝ $1/\sqrt{r}$.
- Used for: high-strength steels, Al/Ti alloys, ceramics, glass, thick sections (plane strain), low-temperature steel.

Criterion: fracture when $K_I = K_{Ic}$ (or $G = G_c$). $K = \sigma Y \sqrt{(\pi a)}$.

4.2 EPFM – Elastic-Plastic Fracture Mechanics

- Applies when **large-scale yielding** occurs at the tip (r_p comparable to a or ligament): ductile metals, low-strength steels, thin sections, high temperatures.
- K no longer unique. Uses **J-integral** (Rice) and **CTOD / δ** (Wells) as fracture parameters.
- Criteria: $J = J_{Ic}$, $\delta = \delta_c$. Resistance curves (J-R, δ-R) describe stable tearing.

LEFM: r_p << a, b
EPFM: r_p ~ a (use J, CTOD)

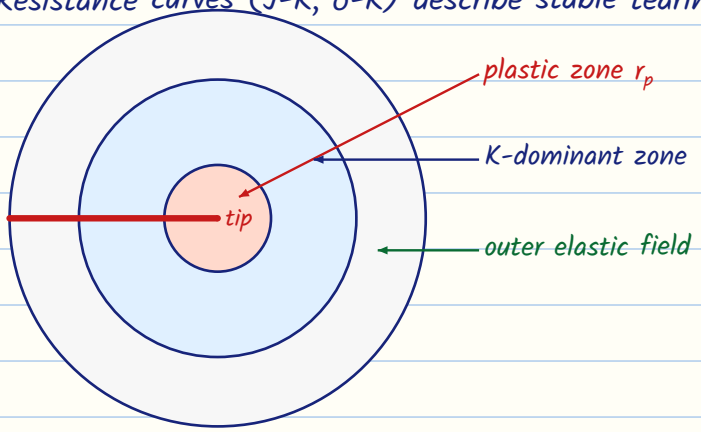


Fig 1.4 Crack-tip zones : K-dominant region vs plastic zone

Feature	LEFM	EPFM
Material behaviour	Linear elastic	Elastic-plastic / non-linear elastic
Plastic zone	Small (SSY), r_p << a	Large, comparable to a, ligament
Parameters	K, G	J, CTOD (δ), tearing modulus T
Fracture criterion	$K = K_{Ic}$, $G = G_c$	$J = J_{Ic}$, $\delta = \delta_c$
Typical materials	High-strength steel, Al, glass, ceramics	Mild steel, stainless steel, Ti, polymers
Standards	ASTM E399	ASTM E1820, BS 7448
Specimen size	Large (plane strain)	Smaller specimens allowed
Mathematics	Simple, closed-form	Needs numerical / FEM, handbooks

Remember

> LEFM ⊂ EPFM : when plasticity → 0, $J \to G = K^2/E'$ ($E' = E$ plane stress; $E/(1-\nu^2)$ plane strain).

> Unit of K : MPa√m (= 10^6 N $m^{-3/2}$); unit of G : N/m (J/m^2); J : N/m; δ : m.

5. Fatigue Crack Growth Model

Fatigue = progressive, localised damage and failure under repeated (cyclic) stress which may be well below the yield strength. About 80-90 % of service failures are fatigue related.

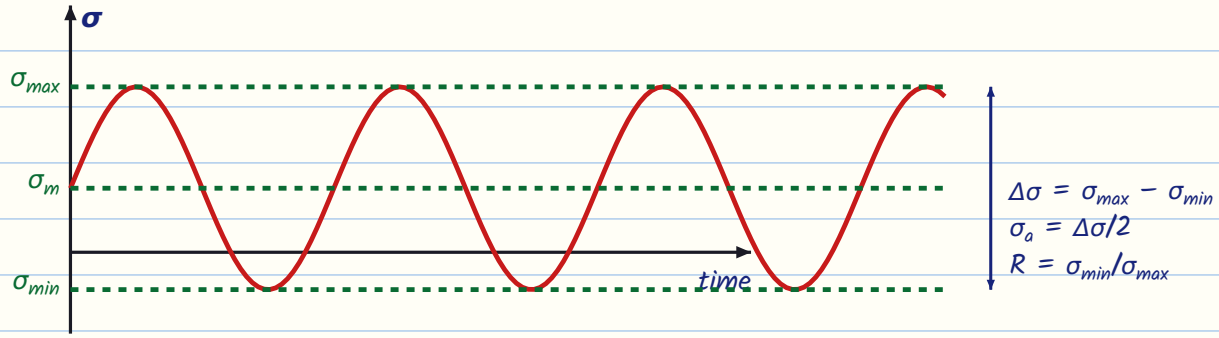


Fig 1.5 Constant-amplitude stress cycle and its parameters

$$\Delta\sigma = \sigma_{max} - \sigma_{min} \quad \sigma_a = \Delta\sigma/2 \quad \sigma_m = (\sigma_{max}+\sigma_{min})/2 \quad R = \sigma_{min}/\sigma_{max}$$

5.1 Three stages of fatigue life

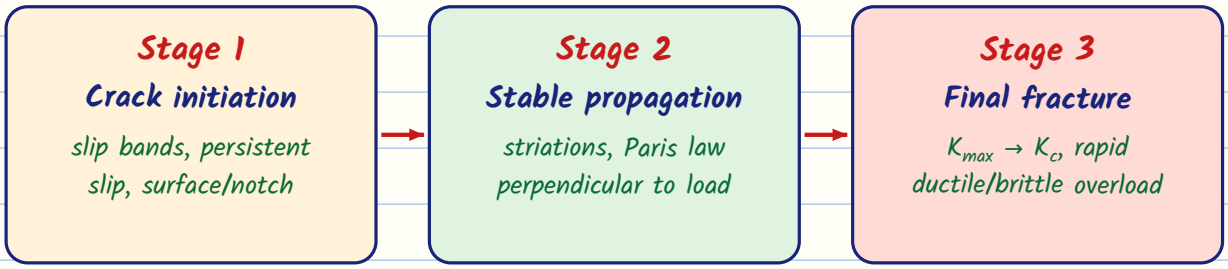


Fig 1.6 Stages of fatigue failure

- **Stage I** (initiation): slip bands, intrusions/extrusions at surface; ~ 90 % of life in high-cycle fatigue; initiates at notches, inclusions.
- **Stage II** (propagation): crack grows ⊥ to max tensile stress; leaves **striations** (one per cycle) and **beach marks** (load changes); governed by ΔK.
- **Stage III** (final fracture): when K_{max} reaches K_c (or net-section yields) – sudden fracture, ductile or brittle.

5.2 S-N approach vs fracture mechanics approach

S-N / strain-life	Fracture mechanics (damage tolerance)
Total life to failure of smooth specimen (initiation dominated)	Crack propagation life from an initial flaw a_0 to a_c
Uses stress amplitude, endurance limit, Goodman/Miner	Uses ΔK, da/dN, Paris law; inspection-based
No crack assumed	Crack assumed (or detected by NDT)

5.3 Fatigue crack growth model (a vs N)

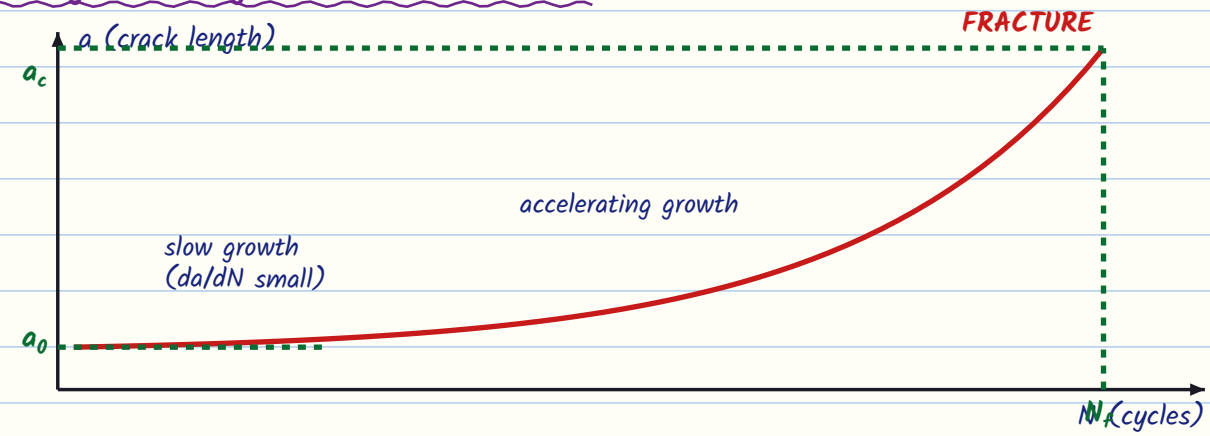


Fig 1.7 Crack length vs number of cycles

Experiment: pre-cracked specimen under constant $\Delta\sigma$; measure a vs N. Slope of the curve = **crack growth rate da/dN**. The rate increases as a grows because $\Delta K = \Delta\sigma\, Y \sqrt{(\pi a)}$ grows with a.

Fundamental hypothesis: the crack-growth rate is controlled by the stress-intensity range:

$$da/dN = f(\Delta K, R) \text{ where } \Delta K = K_{max} - K_{min} = \Delta\sigma \cdot Y \cdot \sqrt{(\pi a)}$$

For $R = 0$, $K_{min} = 0$ so $\Delta K = K_{max}$. (Only the tensile part of the cycle is counted; compressive part ignored.)

5.4 Paris-Erdogan law (basic crack-growth model)

$$da/dN = C\,(\Delta K)^m$$ Paris & Erdogan (1963)

- C, m = material constants (m ≈ 2-4 for metals, ~ 3 for steels; ceramics/polymers m much higher). Units of C depend on m; for da/dN in m/cycle & ΔK in MPa√m.
- Valid in the stable (Region II) growth range only; details & sigmoidal curve in Unit 5.

5.5 Fatigue life by integration

Separate variables and integrate from initial a_0 to critical a_c:

$$N_f = \frac{\int da \;(a_0 \to a_c)}{C\,(\Delta\sigma\, Y \sqrt{(\pi a)})^m}$$

If Y is constant and $m \neq 2$:

$$N_f = \frac{a_c^{\,1-m/2} - a_0^{\,1-m/2}}{C\,(Y\,\Delta\sigma\,\sqrt{\pi})^m\,(1 - m/2)} \qquad m \neq 2$$

For $m = 2$: $N_f = [1 / (C\,\pi\,Y^2\,\Delta\sigma^2)] \cdot \ln(a_c / a_0)$.

Critical crack length from fracture criterion $K_{max} = K_c$:

$$a_c = \frac{1}{\pi}\left(K_c / (Y\,\sigma_{max})\right)^2$$

Steps for a fatigue-life problem (EXAM)

> 1) Get Y (geometry), σ_{max}, $\Delta\sigma$, K_c, C, m, a_0 from data.
> 2) Compute $a_c = (1/\pi)(K_c/(Y\sigma_{max}))^2$.
> 3) Check whether Y varies with a; if constant use closed form, else integrate numerically (stepwise).
> 4) N_f from the formula above; apply safety factor (e.g. $N_f/2$ or /4) to fix inspection interval.

Worked Example 1

A wide steel plate (Y = 1.0) has an initial central crack $2a_0$ = 4 mm and is loaded by $\Delta\sigma$ = 100 MPa (R = 0, σ_{max} = 100 MPa). K_c = 80 MPa√m; C = 6.9×10^{-12} , m = 3 (m/cycle, MPa√m). Find a_c and N_f.

Sol. a_0 = 2 mm = 0.002 m.

$a_c = (1/\pi)(80/100)^2 = 0.64/\pi =$ **0.2037 m** (≈ 204 mm).

$1 - m/2 = -0.5$. $a_0^{-0.5} = 22.36$; $a_c^{-0.5} = 2.215$.

Denominator: $C\,(\Delta\sigma\sqrt{\pi})^3\,(-0.5) = 6.9\times10^{-12} \times (100\times1.7725)^3 \times (-0.5) = 6.9\times10^{-12} \times 5.569\times10^{6} \times (-0.5) = -0.01921$.

Numerator = 2.215 − 22.36 = −20.15.

$N_f = (-20.15)/(-0.01921) \approx 1.05 \times 10^6$ cycles. (Most life is spent while the crack is small!)

Worked Example 2 (critical size)

An edge crack in a large plate (Y = 1.12) carries σ = 250 MPa; K_{Ic} = 55 MPa√m. Critical crack size?

$a_c = (1/\pi)(K_{Ic}/(Y\sigma))^2 = (1/\pi)(55/(1.12\times250))^2 = (1/\pi)(0.1964)^2 =$ **0.01228 m ≈ 12.3 mm**.

Worked Example 3 (edge crack, inspection interval)

Large plate, single edge crack, Y = 1.12, $\Delta\sigma = \sigma_{max}$ = 150 MPa (R = 0). K_c = 60 MPa√m, C = 1×10^{-11}, m = 3. Initial crack a_0 = 1 mm (largest flaw missed by NDT). Find N_f and a safe inspection interval.

$a_c = (1/\pi)(60/(1.12\times150))^2 = (1/\pi)(0.3571)^2 =$ **0.0406 m = 40.6 mm**.

$Y\,\Delta\sigma\,\sqrt{\pi} = 1.12\times150\times1.7725 = 297.8$; $(297.8)^3 = 2.641\times10^7$.

$N_f = (a_c^{-0.5} - a_0^{-0.5}) / [C\,(Y\Delta\sigma\sqrt{\pi})^3\,(-0.5)] = (4.963 - 31.62)/(1\times10^{-11} \times 2.641\times10^7 \times (-0.5)) = (-26.66)/(-1.32\times10^{-4}) \approx$ **2.0×10^5 cycles**.

With a scatter factor of 2 ($N_f/2$) inspect every ≈ 1.0×10^5 cycles. If NDT can find 5 mm

cracks, recompute with a_0 = 5 mm for a more realistic (and longer) schedule.

Worked Example 4 (step-by-step / numerical integration idea)

When Y depends on a, divide the crack range into steps, compute ΔK at the mean a of each step and use ΔN = Δa / (C ΔK^m).

a (mm)	ΔK = Δσ√(πa) (MPa√m)	da/dN = C ΔK^3 (m/cycle)
2	7.93	3.44×10^{-9}
4	11.21	9.72×10^{-9}
8	15.85	2.75×10^{-8}
16	22.42	7.78×10^{-8}
32	31.71	2.20×10^{-7}

(Δσ = 100 MPa, Y = 1, $C = 6.9\times10^{-12}$, m = 3.) Doubling a raises da/dN by $2^{1.5} = 2.83\times$. Cycles to grow 2→4 mm ≈ 3.4×10^{5}; 4→8 mm ≈ 2.4×10^{5}: crack growth **accelerates**.

SA. More on LEFM / EPFM and Fatigue

A. Plane stress and plane strain

- **Thin plate** : free to contract through thickness → $\sigma_z = 0$ → **plane stress**; large plastic zone, shear lips, higher toughness K_c.
- **Thick plate** : through-thickness contraction constrained → $\varepsilon_z = 0$, $\sigma_z = \nu(\sigma_x + \sigma_y)$ → **plane strain**; high triaxiality, small plastic zone, flat fracture, lowest toughness K_{Ic} (a conservative material property).
- Thickness for valid K_{Ic} (ASTM E399): $B, a, (W - a) \geq 2.5\,(K_{Ic} / \sigma_{ys})^2$.
- Plastic-zone size (Irwin): plane stress $r_p = (1/2\pi)(K/\sigma_{ys})^2$; plane strain ≈ one-third of it.

B. Introduction to CTOD and J-integral (EPFM parameters)

- **CTOD (δ)** (Wells, 1961): crack faces blunt and separate before fracture; opening at the original tip = δ. Fracture when $\delta = \delta_c$. In SSY: $\delta = K^2 / (\sigma_{ys} E')$ ($\approx G / \sigma_{ys}$).
- **J-integral** (Rice, 1968): path-independent contour integral = energy-release rate for non-linear elastic body. J = G in linear elasticity. Fracture when $J = J_{Ic}$.
- Experimentally J_{Ic} is measured from a J-R curve (unloading compliance); $K_{Jc} = \sqrt{(E' J_c)}$.

C. Factors affecting fatigue crack growth

Factor	Effect
Stress ratio R / mean stress	Higher R → higher da/dN (K_{max} larger, less closure); threshold ΔK_{th} falls
Overload (tensile)	Plastic zone causes compressive residual stress → crack **retardation** or arrest

Environment	Corrosion fatigue, hydrogen: da/dN increases, lower ΔK_{th}
Frequency & temperature	Low f / high T → time-dependent (creep, oxidation) increases growth
Microstructure, grain size	Coarse grains → better threshold; strength has weak effect in Region II
Thickness	Thin plates (plane stress) show more closure → slower growth

D. Fatigue fracture surface features

- **Beach (clamshell) marks** – macroscopic bands from changes in load or stops/starts; show crack origin and direction (arcs centred on origin).
- **Striations** – microscopic ripples, **one per load cycle** (seen under SEM/TEM). Spacing ≈ da/dN, so striation spacing can be used to estimate growth rate in failure analysis.
- **Ratchet marks** – multiple initiation sites at a notch or corner.
- **Final fast-fracture zone** – rough ductile/brittle region; its size shows how highly the part was stressed.

Cost of fracture

> A US study (Battelle/NBS, 1983) estimated fracture costs ≈ 4 % of GNP (~ $119 billion per year); about a third could be avoided with present knowledge.

> Fracture-control plans (design → material selection → NDT → inspection → repair) are therefore mandatory in aerospace, nuclear, pipelines & offshore.

SB. Glossary & 2-mark Answers

Term	Meaning
Fracture toughness	Resistance of a cracked material to crack extension (K_{Ic}, G_c, J_{Ic})
Stress intensity factor K	Parameter measuring the severity of the crack-tip stress field; $K = \sigma Y\sqrt{(\pi a)}$
Energy release rate G	Energy released per unit new crack area; $G = \pi\sigma^2 a/E$ (plane stress)
Plastic zone	Region ahead of tip where σ exceeds yield stress
Damage tolerance	Ability of a structure to carry load safely with cracks until detected/repaired
ΔK_{th}	Threshold stress-intensity range below which cracks do not grow
Striation	Fatigue marking left by one load cycle
Residual strength	Remaining strength of a cracked structure as a function of crack size

Q: Why do cracks usually start at the surface? Maximum stresses (bending/torsion),

stress raisers, free-surface slip, environment attack.

Q: Is a compressive stress cycle harmful? Crack faces close and carry compression, so only the tensile part contributes to ΔK.

SC. Model Answers (10-mark style)

Q. Explain the fracture-mechanics approach to design.

1. Assume a flaw of size a exists (largest one NDT can miss or has detected).
2. Compute $K = \sigma Y \sqrt{(\pi a)}$ at the critical location for the design load (Y from handbook).
3. Compare with K_{Ic} (plane strain) or K_c. Require $K \leq K_{Ic}$ / FoS.
4. If cyclic loading, compute ΔK and use $da/dN = C \Delta K^m$ to find the number of cycles to reach a_c.
5. Fix inspection interval ≤ N_f / safety factor; repair or replace when a crack is found above $a_{detectable}$.

Residual strength diagram

> Plot of failure stress $\sigma_f = K_c / (Y\sqrt{(\pi a)})$ against crack size a. It falls as a increases: large cracks → low residual strength.

> At $a \to 0$ the curve is limited by σ_{ys} (net-section yield), so the true curve is cut off by a horizontal line at yield/flow stress.

> Fail-safe design keeps the operating stress below the curve for the largest crack that can exist between inspections.

Q. Write notes on Liberty ship failures.

- Welded all-steel cargo ships built in WWII at high speed; all-welded hull let a crack run across several plates (riveted ships arrest cracks at plate edges).
- Steel (low Mn:C ratio, high S) had a high transition temperature, near sea-water temperature (~ 0-5 °C in the North Atlantic) → low toughness.
- Stress raisers: square hatch corners, bad weld details & residual welding stresses.
- Remedies: better steel (Mn:C ≥ 2.5), rounded hatch corners, riveted crack arrester strakes, welding-procedure control, Charpy requirement at service temperature. This work (Irwin, Wells, Naval Research Lab) was the birth of modern fracture mechanics.

Q. Why is K_{Ic} a conservative property?

Plane-strain gives the highest constraint and lowest fracture toughness; thinner sections have $K_c > K_{Ic}$. Using K_{Ic} in design is therefore safe (conservative) for all thicknesses.

Quick MCQs

1) Unit of K_I : (a) MPa (b) **MPa√m** (c) N/m (d) J. Ans: b

2) Mode of loading where crack faces separate normal to the plane : **Mode I**.

3) The Paris law is applicable to : **Region II (stable growth)** of the da/dN–ΔK curve.

4) The crack-growth rate is a function of : **ΔK (and R)**, not of Δσ alone.

5) BCC metals exhibit DBTT; FCC metals (Al, Cu, austenitic steel) generally do **not**.

6. Important Exam Questions (Unit I)

1. Explain with examples three spectacular failures and the lessons learnt for fracture-control design.
2. Differentiate between LEFM and EPFM. State the parameters and applicability of each.
3. Why is the conventional stress-based design inadequate? Explain the three variables of fracture mechanics approach.
4. Describe the three modes of fracture with sketches.
5. Explain the stages of fatigue failure and derive an expression for fatigue life from the Paris law.
6. Compare safe-life, fail-safe and damage-tolerant design philosophies.
7. Explain ductile-brittle transition temperature and the factors that influence it.

Quick revision: $K = \sigma Y\sqrt{(\pi a)}$; $K = K_c \rightarrow$ fracture ; $da/dN = C\ \Delta K^m$; $N_f = (a_c^{1-m/2} - a_0^{1-m/2}) / (C\ (Y\Delta\sigma\sqrt{\pi})^m\ (1 - m/2))$.

UNIT 2

Crack Growth, Fracture Mechanisms, Griffith Theory & Energy Release Rate

fracture mechanisms | theoretical strength | Inglis | Griffith | G, R-curve, compliance

I. Fracture – Basics

Fracture = separation of a body into two or more pieces under stress. It has two stages: (i) **crack initiation** (nucleation) and (ii) **crack propagation**. Types of fracture:

Basis	Types
Plastic deformation	Brittle / Ductile / Quasi-brittle
Crack path	Transgranular (through grains) / Intergranular (along grain boundaries)
Loading	Static overload, fatigue, creep, impact, stress-corrosion, hydrogen-assisted
Crack speed	Stable (slow, needs rising load) / Unstable (fast, self-driven)

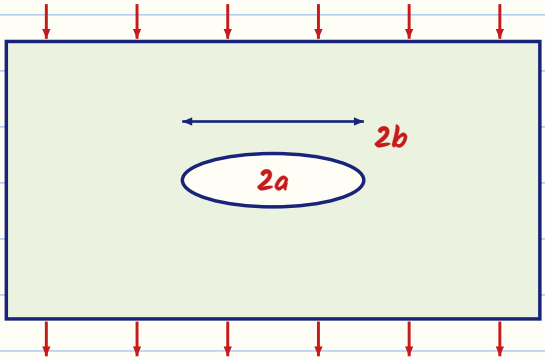
I.1 Theoretical cohesive strength

Atoms separated by a distance x from equilibrium spacing a_0: stress-separation curve approximated by a sine:

$$\sigma = \sigma_{th} \sin(2\pi x/\lambda) \rightarrow \sigma_{th} = \sqrt{(E\,\gamma_s / a_0)} \approx E/10 \text{ (Orowan)}$$

- Derivation: work to separate two planes = $2\gamma_s = \int \sigma\, dx = \sigma_{th}\, \lambda/\pi$; for small x, Hooke $\sigma = E\, x/a_0 = \sigma_{th}\, (2\pi x/\lambda) \Rightarrow \lambda = 2\pi\, a_0\, \sigma_{th} / E$; then $2\gamma_s = \sigma_{th}\, \lambda/\pi = 2\, a_0\, \sigma_{th}^2/E \Rightarrow$ **$\sigma_{th} = \sqrt{(E\gamma_s/a_0)}$**.
- For steel E = 210 GPa → σ_{th} ≈ 20-30 GPa, but measured fracture strength ≈ 0.2-2 GPa, i.e. 10-100 times lower! Glass: E = 70 GPa → σ_{th} ≈ 7-10 GPa; measured ≈ 70 MPa.
- Griffith (1920) explained the discrepancy by **pre-existing microscopic cracks** which amplify stress locally.

2. Inglis Stress Concentration



Inglis (1913):

$\sigma_{max} = \sigma(1 + 2a/b)$

$\rho = b^2/a \rightarrow \sigma_{max} = \sigma(1+2\sqrt{(a/\rho)})$

sharp crack: $\rho \rightarrow 0$, $\sigma_{max} \rightarrow \infty$

(so stress criterion fails!)

Fig 2.1 Elliptical hole in an infinite plate (Inglis 1913)

- Max stress at the end of the major axis: $\sigma_{max} = \sigma\ (1 + 2a/b)$. With root radius $\rho = b^2/a$: $\sigma_{max} = \sigma\ (1 + 2\sqrt{(a/\rho)})$.
- Stress concentration factor $K_t = \sigma_{max}/\sigma = 1 + 2a/b$. Circle ($a = b$): $K_t = 3$.
- For a thin slit/sharp crack, $b \rightarrow 0$ ($\rho \rightarrow 0$) and $\sigma_{max} \rightarrow \infty$: **a stress criterion cannot be used.** Need an energy (Griffith) or K approach.
- For $a/\rho >> 1$: $\sigma_{max} \approx 2\sigma\sqrt{(a/\rho)}$. Real crack tips have $\rho \approx$ atomic radius $\Rightarrow \sigma_{max} \approx \sigma_{th} \Rightarrow$ fracture at $\sigma_f = \sigma_{th}\ \sqrt{(\rho/4a)}$ — which actually matches Griffith's result when $\rho \approx a_0$.

3. Griffith's Theory of Fracture (1921)

Idea: a crack can grow only if the total energy of the system **decreases** (or stays constant) – the energy released from strain energy must be at least the energy needed to create the new surfaces. (First law of thermodynamics: balance.)

Assumptions of Griffith theory

> Material is perfectly elastic, homogeneous, isotropic (ideally brittle, e.g. glass).

> Pre-existing sharp crack of length 2a, through-thickness, in an infinitely wide plate (B = thickness).

> Energy needed for crack growth is only the **surface energy** (γ_s per unit area, two surfaces).

> Crack grows quasi-statically; kinetic energy neglected; applied stress σ is constant & uniform far away.

3.1 Derivation (energy balance)

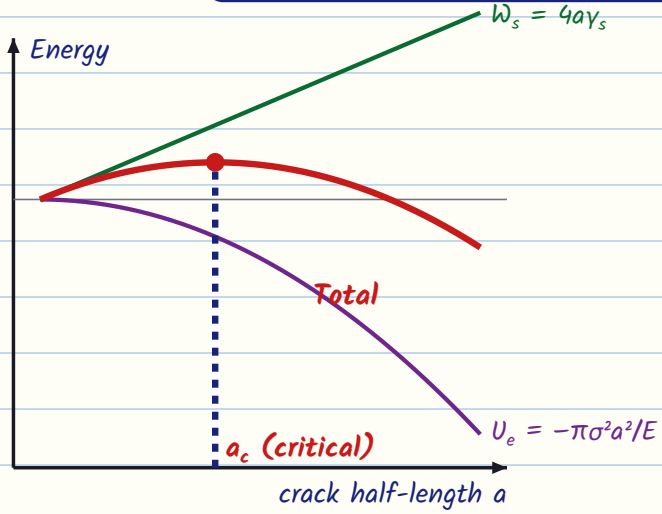
Total energy of the cracked plate: $U = U_0 + U_a + U_\gamma - F$

- U_0 = elastic energy of uncracked plate (constant).
- U_a = change in elastic energy due to crack (Inglis solution) = $-\pi\ \sigma^2\ a^2\ B\ /\ E$ (plane

stress).

- U_γ = surface energy = (2 faces × 2a × B) γ_s = 4 a B γ_s.
- F = work done by external forces; for **fixed grips** F = 0 (no displacement); for fixed load, F = $-2U_a$-type term; result for G is same.

$$U = U_0 - \pi \sigma^2 a^2 B / E + 4 a B \gamma_s$$



$a > a_c$: UNSTABLE
energy released > needed
→ crack runs

$a < a_c$: STABLE
more energy needed to grow

$d(U_e + W_s)/da = 0$
$\Rightarrow \sigma_f = \sqrt{(2E\gamma_s/\pi a)}$

Fig 2.2 Griffith energy balance – the energy barrier at critical crack size

Equilibrium (crack at point of instability): **dU/da = 0**

$$-2\pi \sigma^2 a B / E + 4 B \gamma_s = 0$$

$$\sigma_f = \sqrt{(2 E \gamma_s / (\pi a))}$$ plane stress

Written as $\sigma_f \sqrt{(\pi a)} = \sqrt{(2E\gamma_s)}$ = constant. i.e. for a given material, strength × √(crack size) = const. Here crack size = 2a (central crack); for an edge crack of length a, the same formula applies with a = the edge-crack length (up to a free-surface factor 1.12).

Plane strain form

> Replace E by $E' = E/(1-\nu^2)$: $\sigma_f = \sqrt{(2 E \gamma_s / (\pi (1-\nu^2) a))}$
> Critical crack length: $a_c = 2 E \gamma_s / (\pi \sigma^2)$ (plane stress)
> Condition for growth: $G \geq 2\gamma_s$ where $G = \pi \sigma^2 a / E$.

3.2 Modifications – Irwin & Orowan

- Griffith works only for ideally brittle solids (glass, ceramics, mica). In metals, plastic work γ_p at the crack tip is ~ 10^3 × γ_s (e.g. steel $\gamma_s \approx 1$ J/m², $\gamma_p \approx 10^3$–10^4 J/m²).
- **Irwin–Orowan modification:** replace γ_s by $(\gamma_s + \gamma_p)$:

$$\sigma_f = \sqrt{(2E(\gamma_s + \gamma_p) / (\pi a))} \approx \sqrt{(2E\gamma_p/\pi a)}$$

- Irwin defined $G_c = 2(\gamma_s + \gamma_p)$ = critical energy release rate (fracture toughness in energy terms); fracture when $G = G_c$.

Worked Example 1 (glass)

A glass plate: $E = 70$ GPa, $\gamma_s = 0.5$ J/m², contains a central crack $2a = 2$ μm. Find fracture stress (plane stress).

$a = 1\times10^{-6}$ m. $\sigma_f = \sqrt{(2\ E\ \gamma_s\ /(\pi\ a))} = \sqrt{(2 \times 70\times10^{9} \times 0.5\ /(\pi \times 10^{-6}))} = \sqrt{(2.228\times10^{16})} =$ **1.49×10^{8} Pa ≈ 149 MPa**.

Check: 1 mm crack ($a = 0.5$ mm): $\sigma_f = \sqrt{(7\times10^{10}/(\pi\times5\times10^{-4}))} =$ **6.7 MPa** – shows how strength falls with flaw size.

Worked Example 2 (steel)

Large steel plate, $E = 210$ GPa, $G_c = 2(\gamma_s+\gamma_p) = 20$ kJ/m² . Applied $\sigma = 300$ MPa. Critical half crack length?

From $G = \pi\ \sigma^2\ a\ /\ E = G_c$: $a_c = G_c\ E\ /(\pi\ \sigma^2) = 20000 \times 210\times10^{9}\ /(\pi \times 9\times10^{16}) =$ **0.01485 m ≈ 14.9 mm** (crack length $2a_c \approx 29.7$ mm).

Equivalent $K_c = \sqrt{(E\ G_c)} = \sqrt{(210\times10^{9} \times 20000)} = 64.8$ MPa√m.

4. Energy Release Rate (G)

Definition (Irwin 1956): G = energy available per unit increase in crack area (per unit thickness per unit crack length). It is the **crack driving force**.

$$G = -\ d\Pi/dA = -(1/B)\ d\Pi/da$$

Π = potential energy = $U - F$

- Units: J/m² = N/m. ('Release rate' is not a time rate.)
- For a through crack in infinite plate: $G = \pi\ \sigma^2\ a\ /\ E'$ (E' = E plane stress, $E/(1-\nu^2)$ plane strain).
- Crack extends when $G \geq R$ where R = crack resistance (energy absorbed per unit crack area) = G_c for a flat R-curve. For Griffith, $R = 2\gamma_s$.

4.1 Fixed grip vs fixed load

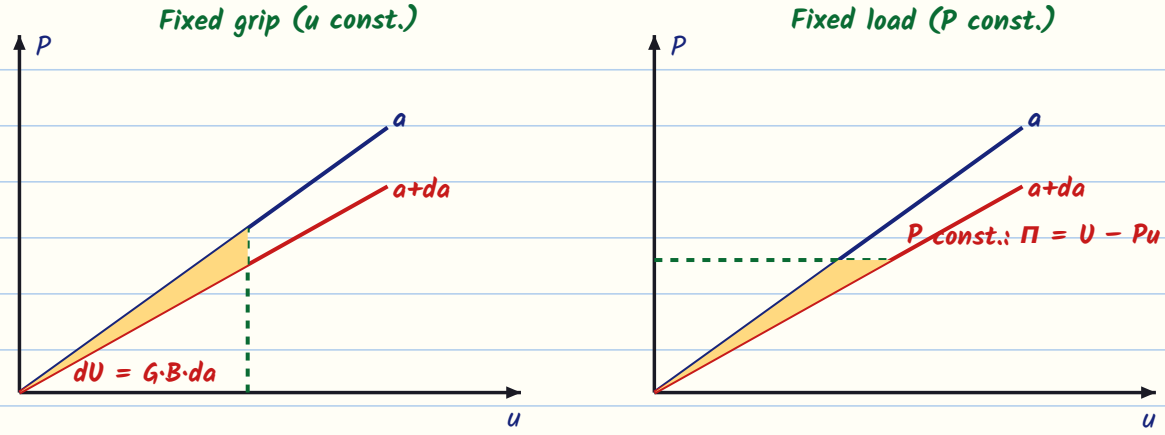


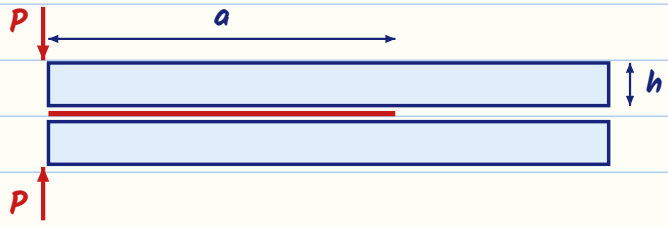
Fig 2.3 Load–displacement response for crack length a and a+da

- **Fixed grip (u const):** crack growth lowers load P; the elastic energy $U = \tfrac{1}{2}Pu$ decreases; $U = u^2/(2C) \Rightarrow G = -(1/B)(\partial U/\partial a)_u = (u^2/2BC^2)(dC/da) = (P^2/2B)(dC/da)$, with compliance $C = u/P$.
- **Fixed load (P const):** U increases by $\tfrac{1}{2}P\,du$ but external work $P\,du$ is done; $\Pi = U - Pu = -\tfrac{1}{2}Pu$ so $G = (1/B)(\partial(\tfrac{1}{2}P\,u)/\partial a)_P = (P^2/2B)(dC/da)$.
- Both give the **same G**:

$$G = (P^2 / 2B) \cdot (dC/da)$$ compliance $C = u/P$

Compliance method is used experimentally: measure C vs a for specimens with different crack lengths, differentiate numerically, get G (and K).

4.2 Example – Double Cantilever Beam (DCB)



Double Cantilever Beam (DCB)

$\delta = 8Pa^3/(E\,B\,h^3)$

$C = \delta/P = 8a^3/(E\,B\,h^3)$

$G = (P^2/2B)(dC/da)$

$G = 12P^2a^2/(E\,B^2\,h^3)$

Fig 2.4 DCB specimen (each arm is a cantilever of depth h)

Each arm: deflection $P\,a^3/(3EI)$ with $I = B\,h^3/12$ → total opening $\delta = 8\,P\,a^3/(E\,B\,h^3)$. $C = 8a^3/(E\,B\,h^3)$. $dC/da = 24\,a^2/(E\,B\,h^3)$.

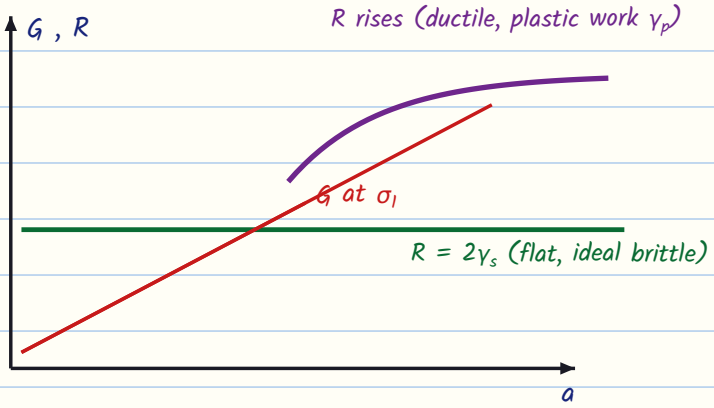
$G = (P^2/2B)(24a^2/(E\,B\,h^3)) = 12\,P^2\,a^2\,/\,(E\,B^2\,h^3)$. Note $G \propto a^2$, so G increases with a at constant P (unstable at fixed load); for fixed displacement δ: $G = 3Eh^3\delta^2/(16a^4)$ decreases with a → **stable** crack growth. (Used in adhesive and composite delamination tests, G_{Ic}.)

Worked Example 3 (DCB)

DCB: $E = 70$ GPa, $h = 10$ mm, $B = 20$ mm, $a = 80$ mm. Find P at which $G = G_c = 400$ J/m².

$P^2 = G\,E\,B^2\,h^3 / (12\,a^2) = 400 \times 70\times10^9 \times (0.02)^2 \times (0.01)^3 / (12 \times (0.08)^2) =$ $400\times70e9\times4e-4\times1e-6/(0.0768) = 1.12\times10^4/0.0768 = 1.458\times10^5 \rightarrow$ **$P \approx 382$ N.**

5. R-curve and Stability of Crack Growth



Stability condition

Growth starts : $G = R$
Stable growth : $dG/da < dR/da$
Unstable : $dG/da \geq dR/da$

Fig 2.5 Crack-driving force G and resistance R

- **Flat R-curve** (ideally brittle; plane strain thick plates): $R = G_c$ constant. Growth starts at $G = R$ and is immediately unstable at fixed load since $dG/da > 0 = dR/da$.
- **Rising R-curve** (ductile; plane stress thin sheets): R increases with Δa because the plastic zone grows. Stable tearing occurs till tangency of G-line with R-curve; instability when **$G = R$ and $dG/da = dR/da$**.
- Stable growth: $dG/da < dR/da$. Used in ASTM ES61 (K-R curve) for thin sheet aluminium alloys.

Relation to Griffith

For Griffith's plate, $G = \pi\,\sigma^2\,a/E \rightarrow$ increases linearly with a at fixed σ. Instability when $G = 2\gamma_s$. For a given crack, increasing σ raises the G-line.

6. Crack Growth and Fracture Mechanisms

6.1 Ductile fracture – micro-void coalescence

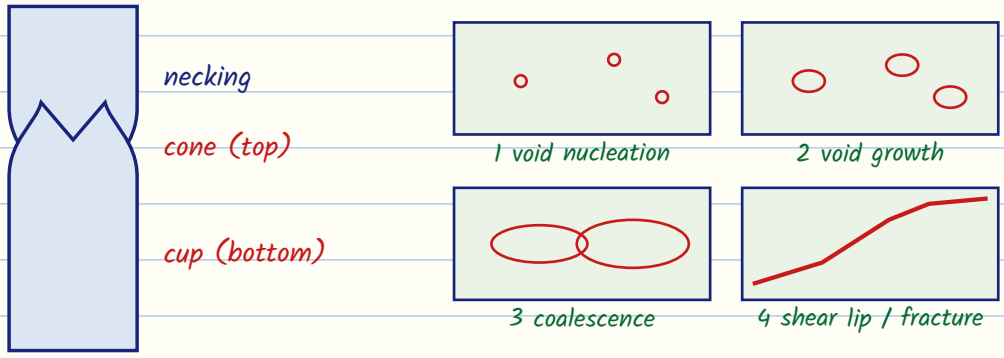


Fig 2.6 Cup-and-cone fracture and its microscopic stages

1. **Necking** begins at UTS: triaxial stress develops at the neck centre.
2. **Void nucleation** at second-phase particles / inclusions (decohesion or cracking).
3. **Void growth** under hydrostatic tension; then **coalescence** by internal necking of ligaments → central flat crack.
4. Crack spreads outward; the outer ring fails by **shear at 45°** (shear lip) → **cup and cone**. Surface shows **dimples**.

6.2 Brittle fracture – cleavage

- **Cleavage**: transgranular separation along low-index crystallographic planes (e.g. {100} in BCC iron) with very little plasticity. Occurs in BCC & HCP metals at low T, high strain rate, triaxial stress; ceramics, glass.
- Fractograph: flat facets; **river patterns** (steps merging in the direction of crack growth, 'tributaries' join 'river'); **chevron (herringbone) marks** point back to the origin; **tongues** from twins.
- Cleavage strength follows a Hall-Petch type relation $\sigma_f \propto d^{-1/2}$ (d = grain size) → fine grain gives higher toughness.

6.3 Intergranular fracture

- Crack follows grain boundaries: segregation of impurities (temper embrittlement – P, Sn, Sb), hydrogen embrittlement, stress-corrosion cracking, creep cavitation, grain-boundary carbides. Surface looks like 'rock candy'.

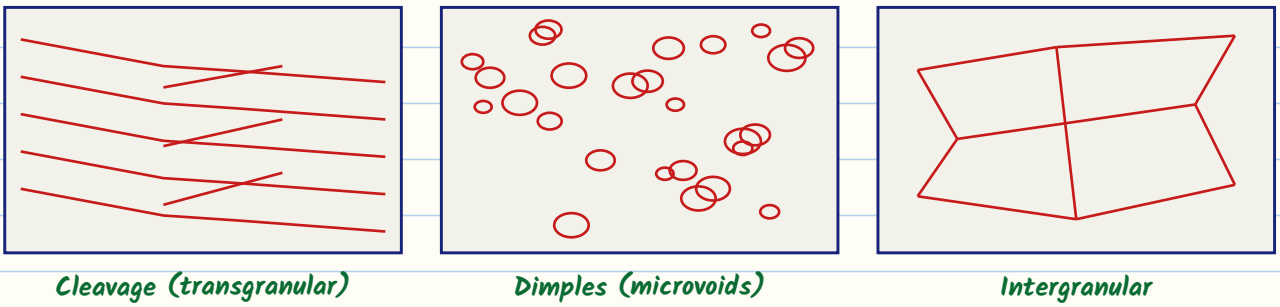


Fig 2.7 Fracture surfaces: cleavage / dimples / intergranular

6.4 Quasi-cleavage & fatigue

- **Quasi-cleavage** : mix of cleavage facets and tear ridges (tempered martensite).
- **Fatigue**: striations (Stage II), beach marks. Mechanism: plastic blunting & re-sharpening of crack tip each cycle (Laird model): tensile half – crack opens & blunts by slip; compressive half – closes, re-sharpens → one striation per cycle.

Mechanism	Typical conditions	Fracture surface	Energy
Cleavage	BCC, low T, high strain-rate, thick	Flat, shiny facets, river marks	Low
Micro-void coalescence	FCC, high T, low rate	Dimples, fibrous, dull	High
Intergranular	Embrittled boundaries, SCC, creep	Faceted grains	Low-med
Fatigue	Cyclic loading	Striations, beach marks	—
Shear rupture	Very pure/soft metals	Chisel point, 100 % necking	High

6.5 Factors that influence fracture

- **Temperature** (DBTT); **strain rate** (higher → brittle); **triaxiality/constraint** (notch, thick section → brittle).
- **Thickness**: K_c falls from the plane-stress value to the plane-strain K_{Ic} as $B \uparrow$; shear lips shrink.
- **Grain size** (fine = tougher), **inclusions**, **heat treatment**, **irradiation**, **environment** (hydrogen, SCC), **residual stress** & **weld defects**.
- **Strength level**: for a given alloy, K_{Ic} usually decreases as σ_{ys} increases (trade-off).

Thickness requirement & plane strain (ASTM E399)

> $B \geq 2.5\,(K_{Ic}/\sigma_{ys})^2$ and $a \geq 2.5\,(K_{Ic}/\sigma_{ys})^2$ and $(W-a) \geq 2.5\,(K_{Ic}/\sigma_{ys})^2$.

> Example: $K_{Ic} = 60$ MPa$\sqrt{m}$, $\sigma_{ys} = 500$ MPa $\Rightarrow B \geq 2.5\,(0.12)^2 = 0.036$ m = 36 mm.

6A. Additional Topics

A. Griffith's experiments & size effect

- Griffith drew glass fibres: strength rose as diameter fell – about **170 MPa** for bulk glass, rising to ~**3.5 GPa** for fibres ≈ 2.5 μm thick (near σ_{th}). Smaller specimens can only contain smaller flaws ⇒ confirms $\sigma_f \propto 1/\sqrt{a}$.
- Hence **whiskers** and glass/carbon fibres are very strong; large brittle components are weak. Statistical (Weibull) strength depends on volume: weakest-link theory.

B. Why surface energy alone is insufficient for metals

Material	γ_s (J/m²)	$G_c = 2(\gamma_s+\gamma_p)$ (J/m²)
Glass	≈ 0.5 – 1	≈ 2 – 10
Alumina (ceramic)	≈ 1	≈ 20 – 40
Mild/structural steel	≈ 1-2	≈ 10^4 – 10^5
Aluminium alloy	≈ 1	≈ 10^4 – 3×10^4

C. Dynamic fracture & crack arrest (Mott)

- Once G exceeds R, extra energy (G – R) goes into **kinetic energy** of material near the moving crack; crack accelerates.
- **Mott / Roberts-Wells:** terminal crack speed ≈ **0.38 √(E/ρ)** (about 0.2-0.4 × shear-wave speed). Steel: ≈ 0.38 × √(210×10⁹/7850) ≈ **1960 m/s**; practical speeds 1000-2000 m/s. Branching occurs near the limiting speed (glass shatters).
- For dynamic loading $K_I(t) \rightarrow K_{Id}$ (dynamic toughness) ≤ K_{Ic} with increased strain rate; **crack-arrest toughness K_{Ia}** < K_{Ic} is the K below which a running crack stops.

D. Sub-critical (environment-assisted) crack growth

- Cracks can grow at $K_I < K_{Ic}$ with time: **stress-corrosion cracking (SCC)**, hydrogen-induced cracking, creep crack growth, corrosion fatigue.
- Threshold **K_{Iscc}** : below it no growth in that environment (e.g. high-strength steel in sea water: K_{Iscc} ≈ 0.2-0.4 K_{Ic}). Growth rate da/dt vs K shows 3 regions: (I) steep rise, (II) plateau (diffusion controlled), (III) $K \rightarrow K_{Ic}$.
- Design rule for sustained loads: keep $K_I < K_{Iscc}$. Time to failure from K_i to K_{Ic} is found by integrating $da/dt = f(K)$.

E. Fractography (failure analysis tools)

- **Macro-fractography** (eye, stereo-microscope): origin, beach marks, chevron, shear lips, direction of crack travel.
- **SEM** (micro-fractography): dimples, facets, striations (spacing → da/dN), intergranular facets; EDS for corrosion products. **TEM** replicas for fine striations.
- **Failure-analysis steps:** collect history → preserve fracture faces → visual → NDT → chemical/ mechanical tests → fractography → stress analysis (FM) → report & remedy.

Solved Numerical Problems

P1 (Inglis). Elliptical hole a = 5 mm, b = 0.5 mm in plate with σ = 100 MPa. σ_{max} = 100 (1 + 2×5/0.5) = **2100 MPa** (K_t = 21); root radius ρ = b^2/a = 0.05 mm.

P2 (ceramic). Alumina, E = 380 GPa, γ_s = 1 J/m². For σ = 100 MPa the critical half-crack length $a_c = 2E\gamma_s/(\pi\sigma^2) = 2\times380\times10^9\times1/(\pi\times10^{16}) =$ **2.4×10^{-5} m = 24 µm** (so grain-size

flaws control strength).

P3 (plane strain). Steel $E = 210$ GPa, $\nu = 0.3$, $G_c = 10$ kJ/m², $a = 10$ mm. Plane stress: $\sigma_f = \sqrt{(E\,G_c/(\pi\,a))} =$ **258 MPa**; plane strain: $\sigma_f = \sqrt{(E\,G_c\,/(\pi(1-\nu^2)a))} =$ **271 MPa** (the plane-strain value is higher for the same G_c, but real G_c is lower in plane strain).

P4 (compliance). At $P = 2$ kN, $B = 10$ mm, compliance rises from $C = 1.00\times10^{-7}$ to 1.10×10^{-7} m/N when a goes 50 → 52 mm. $dC/da \approx 5\times10^{-5}$ (m/N)/m. $G = (P^2/2B)(dC/da) = (4\times10^6/0.02)\times5\times10^{-5} =$ **1.0×10^4 J/m²**; $K = \sqrt{(E\,G)} =$ **45.8 MPa√m** ($E = 210$ GPa, plane stress).

P5 (thickness). Alloy $\sigma_{ys} = 450$ MPa, $K_{Ic} = 80$ MPa√m: $B \geq 2.5\,(80/450)^2 = 0.079$ m = **79 mm** → a thin sheet will not give a valid K_{Ic}; use K_c / R-curve or EPFM.

6B. G for Standard Cases & Summary

Configuration	Energy release rate G
Central crack 2a, infinite plate, σ	$\pi\,\sigma^2\,a\,/\,E'$
Single edge crack a, semi-infinite plate	$1.12^2\,\pi\,\sigma^2\,a\,/\,E' \approx 1.25\,\pi\,\sigma^2\,a/E'$
DCB (arm depth h), load P	$12\,P^2\,a^2\,/\,(E\,B^2\,h^3)$
Infinite strip (height 2h), fixed grip, strain ε	$G = \sigma^2\,h\,/\,E = E\,\varepsilon^2\,h$ (independent of a)
Penny-shaped crack radius a, σ	$4\,(1-\nu^2)\,\sigma^2\,a\,/\,(\pi\,E)$
Any body (compliance method)	$(P^2/2B)\,dC/da$

Griffith vs Irwin vs Orowan (revision)

> **Griffith (1921):** energy balance, only surface energy, ideally brittle: $\sigma_f = \sqrt{(2E\gamma_s/\pi a)}$.

> **Irwin (1948):** includes plastic work: $G_c = 2(\gamma_s+\gamma_p)$; introduced $G = -d\Pi/dA$ (1956) and K (1957); $G = K^2/E'$.

> **Orowan (1949):** independently suggested $\gamma_p \gg \gamma_s$ for metals; $\sigma_f = \sqrt{(2E(\gamma_s+\gamma_p)/\pi a)}$.

Worked Example 4 (strip, fixed grip)

Thin strip of height $2h = 100$ mm, $E = 70$ GPa, $G_c = 20$ kJ/m². Gripped at fixed displacement. Find the stress/strain at which a long edge crack runs.

$G = \sigma^2\,h\,/\,E = G_c \Rightarrow \sigma_c = \sqrt{(G_c\,E\,/\,h)} = \sqrt{(2\times10^4 \times 7\times10^{10}\,/\,0.05)} =$ **167 MPa** ; $\varepsilon_c = \sigma_c/E =$ **0.0024** (0.24 %).

G is independent of crack length ⇒ once started, the crack grows at constant G (steady-state propagation).

Worked Example 5 (penny-shaped crack)

Buried penny-shaped crack, radius $a = 5$ mm, $\sigma = 150$ MPa, steel ($E = 210$ GPa, $\nu = 0.3$). $K_I = 2\sigma\sqrt{(a/\pi)} = 2\times150\times\sqrt{(0.005/\pi)} =$ **12.0 MPa√m**. $G = (1-\nu^2)K^2/E = 0.91\times(11.97\times10^6)^2/(210\times10^9) =$ **621 J/m²**.

Key Points to Remember

- A crack grows when the **rate of energy release G ≥ rate of energy absorption R** (necessary condition – Griffith criterion is necessary, not sufficient, since the energy must also be transferable at the tip by stress).
- G is a **global energy** quantity; K is a **local stress-field** quantity; they are linked by $G = K^2/E'$ (Unit III).
- G depends on boundary conditions only through the geometry/compliance, **not** on whether grips are fixed or the load is fixed.
- Fixed-grip loading generally gives more stable crack growth (G falls as a grows for a DCB); dead-weight loading is the most unstable.

Unsolved Practice Problems (with answers)

1. Glass, $E = 70$ GPa, $\gamma_s = 0.6$ J/m²; central crack of length $2a = 0.2$ mm. Find σ_f. Ans: **≈ 16.4 MPa**
2. Steel, $E = 210$ GPa, $G_c = 15$ kJ/m², $\sigma = 200$ MPa. Find a_c. Ans: **≈ 25.1 mm**
3. DCB: $P = 300$ N, $a = 60$ mm, $h = 8$ mm, $B = 15$ mm, $E = 70$ GPa. Find G and K. Ans: **G ≈ 482 J/m²**, $K = \sqrt{(EG)} \approx$ **5.8 MPa√m**
4. Show that for Griffith's plate $\sigma_f^2\, a =$ const. and plot σ_f vs a (hyperbola-type curve; log–log slope = −1/2).

2-mark Short Answers

Q. Define surface energy. Energy required to create unit area of new free surface (J/m²).

Q. State Griffith criterion. A crack propagates if the elastic energy released by an increment of growth ≥ the surface energy required.

Q. What is compliance? C = displacement/load = u/P; reciprocal of stiffness; depends on crack length.

Q. Difference between G and K? G = energy approach (global), K = stress-field approach (local); $G = K^2/E'$.

Q. What is an R-curve? Plot of crack resistance R vs crack extension Δa; flat for brittle, rising for ductile materials.

Q. Give two features of cleavage. Flat bright facets, river patterns, low energy, along crystal planes.

Q. What is a dimple rupture? Fracture by void nucleation, growth and coalescence –

cup-and-cone surfaces have dimples.

7. Exam Questions & Quick Answers

1. Derive Griffith's criterion for brittle fracture. State the assumptions and the modification for ductile materials.
2. Define energy release rate. Show that $G = (P^2/2B)\, dC/da$ and that it is the same for fixed load and fixed grip.
3. Explain the R-curve and conditions for stable and unstable crack growth.
4. Describe the mechanisms of ductile and brittle fracture with sketches.
5. Why does the Inglis solution fail to give a fracture criterion for sharp cracks?
6. Compute G and critical load for a DCB specimen (numerical).

Formula sheet – Unit II

> $\sigma_{th} = \sqrt{(E\gamma_s/a_0)} \approx E/10$ | $\sigma_{max} = \sigma(1+2\sqrt{(a/\rho)})$

> Griffith: $\sigma_f = \sqrt{(2E\gamma_s/\pi a)}$; plane strain $E \to E/(1-\nu^2)$; ductile: $\gamma_s \to \gamma_s+\gamma_p$

> $G = \pi\sigma^2 a/E'$; $G = (P^2/2B)(dC/da)$; $G_c = R = 2(\gamma_s+\gamma_p)$

> DCB: $G = 12P^2a^2/(E\, B^2\, h^3)$; Instability: $G = R$ and $dG/da \geq dR/da$

UNIT 3

Theory of Elasticity, Westergaard Solution (Mode I) & the K–G Relation

stress & strain review | Airy function | Westergaard | stresses & displacements | $G = K^2/E'$

1. Review of Theory of Elasticity

LEFM is built on the **linear theory of elasticity**; the crack-tip stresses are obtained by solving the elasticity equations with the crack-face boundary conditions.

1.1 Stress and equilibrium

- Stress tensor σ_{ij} $(i, j = x, y, z)$: 3 normal $(\sigma_x, \sigma_y, \sigma_z)$ + 3 shear $(\tau_{xy} = \tau_{yx}, \tau_{yz}, \tau_{zx})$ components.
- **Equilibrium** (no body force), 2-D:

$$\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y = 0 \; ; \; \partial\tau_{xy}/\partial x + \partial\sigma_y/\partial y = 0$$

- Principal stresses (2-D): $\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{[\,((\sigma_x - \sigma_y)/2)^2 + \tau_{xy}^2\,]}$; max shear $\tau_{max} = (\sigma_1 - \sigma_2)/2$.

1.2 Strain & compatibility

$$\varepsilon_x = \partial u/\partial x \quad \varepsilon_y = \partial v/\partial y \quad \gamma_{xy} = \partial u/\partial y + \partial v/\partial x$$

- u, v = displacements in x, y. For a single-valued continuous displacement field the strains must satisfy **compatibility**:

$$\partial^2\varepsilon_x/\partial y^2 + \partial^2\varepsilon_y/\partial x^2 = \partial^2\gamma_{xy}/\partial x\partial y$$

1.3 Hooke's law, plane stress & plane strain

- Isotropic: $\varepsilon_x = [\sigma_x - \nu(\sigma_y + \sigma_z)]/E$; $\varepsilon_y = [\sigma_y - \nu(\sigma_x + \sigma_z)]/E$; $\gamma_{xy} = \tau_{xy}/\mu$; shear modulus $\mu = E / 2(1 + \nu)$.

	Plane stress (thin plate)	Plane strain (thick plate)
Assumption	$\sigma_z = \tau_{xz} = \tau_{yz} = 0$	$\varepsilon_z = \gamma_{xz} = \gamma_{yz} = 0$
Out-of-plane	$\varepsilon_z = -\nu(\sigma_x + \sigma_y)/E$	$\sigma_z = \nu(\sigma_x + \sigma_y)$
Kolosov constant κ	$(3 - \nu)/(1 + \nu)$	$3 - 4\nu$
Modulus E'	E	$E/(1 - \nu^2)$

1.4 Airy stress function φ(x, y)

Choose a scalar φ that **automatically satisfies equilibrium**:

$$\sigma_x = \partial^2\varphi/\partial y^2 \quad \sigma_y = \partial^2\varphi/\partial x^2 \quad \tau_{xy} = -\partial^2\varphi/\partial x\partial y$$

Substituting in compatibility (with Hooke's law) gives the **bi-harmonic equation**:

$$\nabla^4\varphi = \partial^4\varphi/\partial x^4 + 2\,\partial^4\varphi/\partial x^2\partial y^2 + \partial^4\varphi/\partial y^4 = 0$$

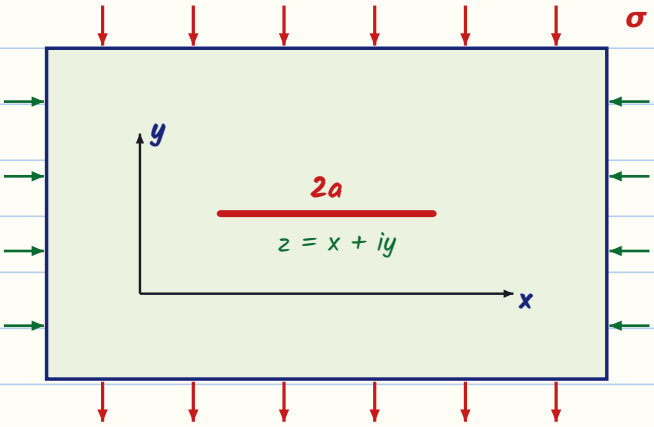
same for plane stress & strain

- Solve $\nabla^4\varphi = 0$ with the boundary conditions → stresses; strains from Hooke; displacements by integration.
- In polar form: $\sigma_r = (1/r)\,\partial\varphi/\partial r + (1/r^2)\,\partial^2\varphi/\partial\theta^2$, $\sigma_\theta = \partial^2\varphi/\partial r^2$, $\tau_r\theta = -\partial/\partial r[(1/r)\,\partial\varphi/\partial\theta]$.

1.5 Complex-variable method

- Let $z = x + iy = r\,e^{i\theta}$. A function $f(z) = \alpha + i\beta$ is **analytic** if Cauchy–Riemann conditions $\partial\alpha/\partial x = \partial\beta/\partial y$, $\partial\alpha/\partial y = -\partial\beta/\partial x$ hold. Then α and β are harmonic: $\nabla^2\alpha = \nabla^2\beta = 0$.
- Hence $\varphi = \mathrm{Re}[\,\mathrm{conj}(z)\,\psi(z) + \chi(z)\,]$ (Goursat) satisfies $\nabla^4\varphi = 0$ — any two analytic functions give a valid stress field (Kolosov–Muskhelishvili). Westergaard (1939) simplified this to **one** function Z(z) for symmetric (Mode I) problems.

2. Westergaard Solution – Mode I



Centre crack, infinite plate, biaxial remote stress σ

$$Z_I(z) = \sigma\, z \,/\, \sqrt{(z^2 - a^2)}$$

B.C.: $\sigma_y = 0$ on $|x| < a$

$\sigma_x = \sigma_y = \sigma$ as $|z| \to \infty$

Fig 3.1 Westergaard problem: crack 2a in an infinite plate with biaxial σ

Westergaard stress function (Mode I)

> Airy function: $\varphi = \mathrm{Re}\, Z_2 + y\, \mathrm{Im}\, Z_1$ where $Z_1 = \int Z\, dz$, $Z_2 = \int Z_1\, dz$ and $Z(z)$ is analytic.

> Stresses: $\sigma_x = \mathrm{Re}\, Z - y\, \mathrm{Im}\, Z'$ $\sigma_y = \mathrm{Re}\, Z + y\, \mathrm{Im}\, Z'$ $\tau_{xy} = -y\, \mathrm{Re}\, Z'$ ($Z' = dZ/dz$)

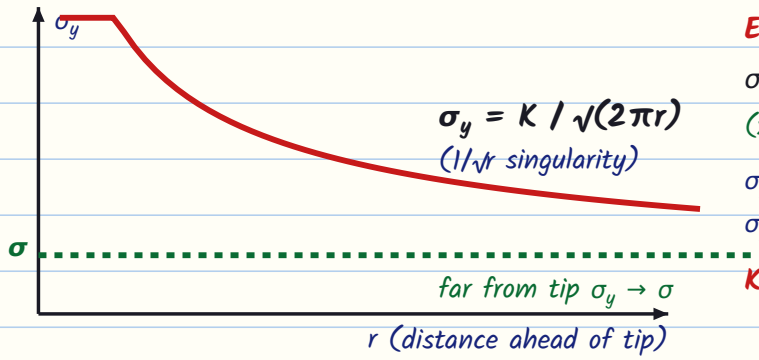
> Displacements (2μ factor): $2\mu u = ((\kappa-1)/2)\, \mathrm{Re}\, Z_1 - y\, \mathrm{Im}\, Z$; $2\mu v = ((\kappa+1)/2)\, \mathrm{Im}\, Z_1 - y\, \mathrm{Re}\, Z$

2.1 Check of boundary conditions for a crack

For a crack on $-a \le x \le a$ ($y = 0$) with remote biaxial tension σ, take

$$Z_1(z) = \sigma z / \sqrt{(z^2 - a^2)}$$

1. Far field ($|z| \to \infty$): $Z \to \sigma$, $Z' \to 0 \Rightarrow \sigma_x = \sigma_y = \sigma$, $\tau_{xy} = 0$ ✔.
2. On crack faces $y = 0$, $|x| < a$: $\sqrt{(x^2 - a^2)}$ is imaginary $\Rightarrow Z$ is purely imaginary $\Rightarrow \sigma_y = \mathrm{Re}\, Z = 0$ and $\tau_{xy} = 0$ ✔ (traction-free).
3. Ahead of crack $y = 0$, $x > a$: Z real $\Rightarrow$ **$\sigma_y = \sigma x / \sqrt{(x^2 - a^2)}$** $\to$ singular at $x = a$.



Exact (Westergaard):

$\sigma_y = \sigma x / \sqrt{(x^2 - a^2)}$

(x = distance from crack centre)

$\sigma_y \to \infty$ as $x \to a$: singular

$\sigma_y \to \sigma$ as $x \to \infty$

K-field valid for $r \ll a$

Fig 3.2 Stress ahead of the crack tip ($\theta = 0$)

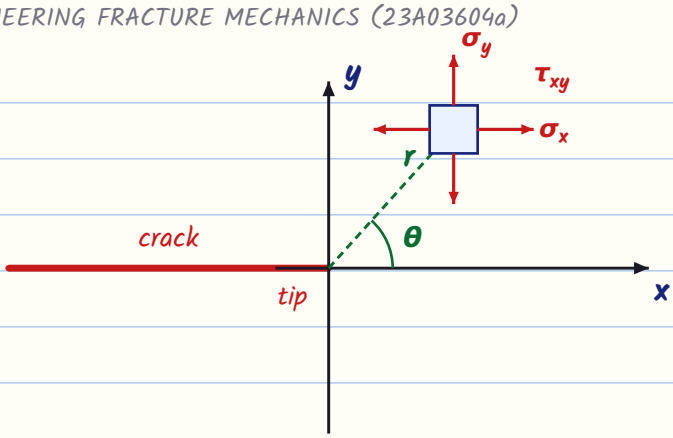
Remark (uniaxial loading): if $\sigma_x = 0$ at infinity the above solution must be corrected by a uniform $\sigma_x = -\sigma$ (**T-stress**). It does not change σ_y, τ_{xy} or K, only σ_x (matters in Unit IV).

2.2 Stress field near the tip – K defined

Shift the origin to the right tip: $z = a + \zeta$, $\zeta = r e^{i\theta}$, $r \ll a$. Then $z^2 - a^2 = \zeta(2a + \zeta) \approx 2a\zeta$ and

$$Z_1 \approx \sigma a / \sqrt{(2a\zeta)} = \sigma\sqrt{(\pi a)} / \sqrt{(2\pi\zeta)} = K_I / \sqrt{(2\pi\zeta)}$$

$K_I = \sigma\sqrt{(\pi a)}$



Polar coordinates at the tip:

$x = r\cos\theta$, $y = r\sin\theta$

$\theta = 0$: line ahead of the crack

$\theta = \pm\pi$: crack faces (free)

Origin shifted from crack centre to the crack tip: $z = a + \zeta$

$\zeta = r\,e^{i\theta}$

Fig 3.3 Crack-tip coordinate system and stress components

Differentiate & separate real and imaginary parts: $Z = K/\sqrt{(2\pi r)} \cdot e^{-i\theta/2}$, $Z' = -K/(2\sqrt{(2\pi)}\, r^{3/2}) \cdot e^{-3i\theta/2}$. Using $\sigma_y = \mathrm{Re}\, Z + y\, \mathrm{Im}\, Z'$ etc. ($y = r\sin\theta$):

$$\sigma_x = K_I/\sqrt{(2\pi r)} \cdot \cos(\theta/2)\,[\,1 - \sin(\theta/2)\sin(3\theta/2)\,]$$

$$\sigma_y = K_I/\sqrt{(2\pi r)} \cdot \cos(\theta/2)\,[\,1 + \sin(\theta/2)\sin(3\theta/2)\,]$$

$$\tau_{xy} = K_I/\sqrt{(2\pi r)} \cdot \cos(\theta/2)\sin(\theta/2)\cos(3\theta/2)$$

Plane strain out-of-plane stress

> $\sigma_z = \nu\,(\sigma_x + \sigma_y) = 2\,\nu\,K_I\cos(\theta/2) / \sqrt{(2\pi r)}$ (plane strain); $\sigma_z = 0$ (plane stress).

> Principal stresses: $\sigma_{1,2} = K/\sqrt{(2\pi r)}\cos(\theta/2)\,[1 \pm \sin(\theta/2)]$; $\sigma_3 = \nu(\sigma_1+\sigma_2)$ (plane strain).

2.3 Properties of the K-field

- **Singularity** $\propto r^{-1/2}$: stresses $\to \infty$ as $r \to 0$; only the *intensity* K is finite. (Real material yields at the tip – plastic zone.)
- On the crack line $\theta = 0$: $\sigma_x = \sigma_y = K/\sqrt{(2\pi r)}$, $\tau_{xy} = 0$ (pure hydrostatic-like tension, max triaxiality in plane strain).
- **Definition:** $K_I = \lim_{r\to 0} \sigma_y(r, \theta = 0)\ \sqrt{(2\pi r)}$. Unit: MPa$\sqrt{m}$ (or N/m$^{3/2}$).
- The shape of the field ($f_{ij}(\theta)$) is universal; **K scales its magnitude** – same K ⇒ same crack-tip state for any geometry (basis of similitude, K_{Ic} transferable from specimen to structure).
- K depends linearly on load, and on crack size & geometry: **$K = \sigma\, Y\, \sqrt{(\pi a)}$**. Superposition of K's valid for the **same mode**.
- K-dominance zone: valid for r small compared with a, ligament and thickness ($r <$ ~ a/10 to a/50); beyond that, higher-order terms/T-stress matter.

Worked Example 1 (stresses near the tip)

K_I = 40 MPa√m, plane strain, ν = 0.3. At r = 2 mm find σ_x, σ_y, τ_{xy}, σ_z at θ = 0 and θ = 90°.

$K/\sqrt{(2\pi r)} = 40/\sqrt{(2\pi\times0.002)} = 40/0.1121 =$ **356.8 MPa**.

θ = 0 : $\sigma_x = \sigma_y =$ **356.8 MPa**, $\tau_{xy} = 0$, $\sigma_z = \nu(\sigma_x+\sigma_y) = 0.3\times713.6 =$ **214 MPa**.

θ = 90° : cos45° = sin45° = 0.7071, sin135° = 0.7071, cos135° = −0.7071 → σ_x = 356.8×0.7071×(1−0.5) = **126 MPa** ; σ_y = 356.8×0.7071×1.5 = **378 MPa** ; τ_{xy} = 356.8×0.5×(−0.7071) = **−126 MPa** ; σ_z = 0.3×(126+378) = **151 MPa**.

3. Displacements and Crack Opening

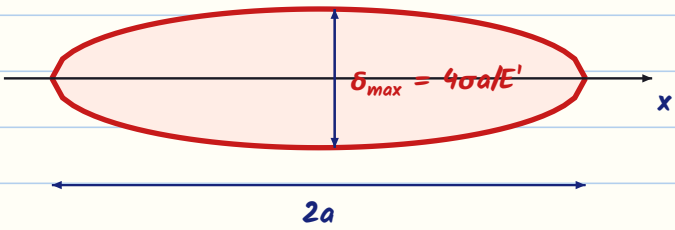
Integrating Z: $Z_I = \int Z\, d\zeta = K\sqrt{(2\zeta/\pi)}$. Then near the tip (κ = Kolosov constant):

$$u = (K_I/2\mu)\sqrt{(r/2\pi)}\cos(\theta/2)\,[\,\kappa - 1 + 2\sin^2(\theta/2)\,]$$

$$v = (K_I/2\mu)\sqrt{(r/2\pi)}\sin(\theta/2)\,[\,\kappa + 1 - 2\cos^2(\theta/2)\,]$$

- Displacements ∝ √r (finite at tip, strain energy finite). On the crack faces θ = ±π: **$v = \pm(\kappa+1)K_I/(2\mu)\cdot\sqrt{(r/2\pi)}$** , $u = \pm(\kappa-1)K_I/(2\mu)\cdot\sqrt{(r/2\pi)}$.
- Full-field: using $Z_I = \sigma\sqrt{(z^2-a^2)}$ the face opening of the Griffith crack is an **ellipse**:

$$v(x) = (2\sigma/E')\sqrt{(a^2-x^2)} \Rightarrow \delta_{max}\ (\text{centre, total}) = 4\sigma a/E'$$



Crack-face displacement:

$v(x) = (2\sigma/E')\sqrt{(a^2-x^2)}$

(each face, using $\int Z\,dz = \sigma\sqrt{(z^2-a^2)}$)

Shape = ELLIPSE (not a wedge)

$E' = E$ (plane stress)

$E' = E/(1-\nu^2)$ (plane strain)

Fig 3.4 Crack opening displacement profile (elliptical)

- CTOD concept: opening at distance r behind the tip $\delta = 2v = (\kappa+1)K/\mu\cdot\sqrt{(r/2\pi)}$ → used by Irwin to define δ_t (Unit IV).
- Strain energy density near tip ∝ 1/r (integrable singularity) – the basis of Sih's S-criterion.

Worked Example 2 (crack face opening)

K_I = 30 MPa√m, E = 70 GPa, ν = 0.33 (plane strain). Find v at r = 1 mm behind tip on the upper face.

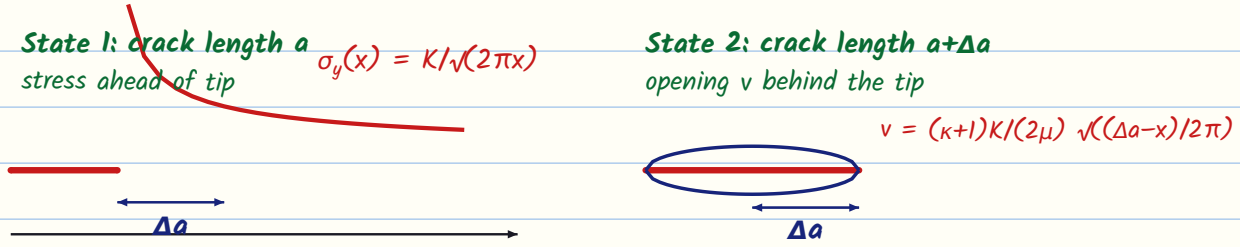
$\mu = E/2(1+\nu)$ = 26.3 GPa ; $\kappa = 3 - 4\nu$ = 1.68 ; $v = (K/2\mu)\sqrt{(r/2\pi)}(\kappa+1)$ = $(30\times10^6/(2\times26.3\times10^9))\times\sqrt{(10^{-3}/2\pi)}\times2.68$ = 5.70×10^{-4} × 0.01262 × 2.68 = **1.93×10^{-5} m ≈ 19.3 µm**.

Worked Example 3 (central crack)

Steel plate (E = 210 GPa, plane stress), $2a$ = 40 mm, σ = 150 MPa. $K = \sigma\sqrt{(\pi a)}$ = $150\sqrt{(\pi\times0.02)}$ = **37.6 MPa√m**; centre opening $\delta_{max} = 4\sigma a/E$ = $4\times150\times10^6\times0.02/(210\times10^9)$ = **57 µm**.

4. Relationship Between K and G

Irwin's crack closure integral: G equals the work needed to close the crack by a small length Δa (= work released when it extends by Δa). Consider two states: (1) crack a with stress $\sigma_y(x)$ ahead of tip; (2) crack $a + \Delta a$ with opening v behind the new tip.



$$G = \lim (1/\Delta a) \int_0^{\Delta a} \sigma_y(x) \cdot v(\Delta a - x)\, dx = \text{work to close the crack by } \Delta a$$

Fig 3.5 Irwin's crack-closure integral

Work to close = ½ ∫ $\sigma_y \cdot (2v)\, dx$ = ∫ $\sigma_y\, v\, dx$ (upper + lower face, linear spring, factor ½):

$$G = \lim_{\Delta a \to 0} (1/\Delta a) \int_0^{\Delta a} \sigma_y(x)\, v(\Delta a - x)\, dx$$

Substitute $\sigma_y = K/\sqrt{(2\pi x)}$ and $v(\Delta a - x) = (\kappa+1)\, K/(2\mu) \cdot \sqrt{((\Delta a - x)/2\pi)}$:

$$G = (\kappa+1)\, K^2 / (4\pi\, \mu\, \Delta a) \int_0^{\Delta a} \sqrt{((\Delta a - x)/x)}\, dx = (\kappa+1)\, K^2 / (4\pi\, \mu\, \Delta a) \cdot (\pi\, \Delta a / 2)$$

$$G = (\kappa + 1)\, K^2 / (8\, \mu)$$

- Plane stress: $\kappa + 1 = 4/(1+\nu)$, $\mu = E/2(1+\nu) \Rightarrow G = K^2/E$.
- Plane strain: $\kappa + 1 = 4(1-\nu)$, $\Rightarrow G = (1-\nu^2)\, K^2/E$.

K – G relation (very important)

> $G = K_I^2 / E'$ with $E' = E$ (plane stress) ; $E' = E/(1-\nu^2)$ (plane strain).

> Mixed mode (I, II, III): $G = (K_I^2 + K_{II}^2)/E' + K_{III}^2/(2\mu)$.

> Fracture toughness link: $K_{Ic} = \sqrt{(E' G_{Ic})}$; $G_c = K_c^2/E'$.

> Check with Griffith: $K = \sigma\sqrt{(\pi a)} \Rightarrow G = \pi \sigma^2 a/E$ ✓ (same as the energy approach).

> ∴ Energy criterion ($G = G_c$) and stress-field criterion ($K = K_c$) are **equivalent** in LEFM.

Closure integral also shows G is **always positive** and additive only between modes: $G_{total} = G_I + G_{II} + G_{III}$ (no cross terms because work done by σ_y on v_{II} etc. is zero).

Worked Example 4

Crack $2a = 50$ mm in a wide plate under $\sigma = 100$ MPa ($E = 70$ GPa, plane stress). $K = 100\sqrt{(\pi \times 0.025)} = 28.0$ **MPa√m** ; $G = K^2/E = (28.0\times10^6)^2/(70\times10^9) = 1.12\times10^4$ **J/m² = 11.2 kJ/m²**.

Worked Example 5

Steel $K_{Ic} = 50$ MPa√m, $E = 210$ GPa, $\nu = 0.3$. Plane-strain $G_{Ic} = (1-\nu^2)K_{Ic}^2/E = 0.91\times(50\times10^6)^2/(210\times10^9) =$ **10.8 kJ/m²**. For $\sigma = 250$ MPa critical half-crack: $a_c = (1/\pi)(K_{Ic}/\sigma)^2 = (1/\pi)(0.2)^2 =$ **12.7 mm**.

Worked Example 6 (ratio of toughness)

Plane-stress $K_c = 100$ MPa√m in thin sheet vs plane-strain $K_{Ic} = 40$ MPa√m of the same alloy ⇒ G_c ratio ≈ $(100/40)^2/(1-\nu^2)$ ≈ 6.9 (E same): thin sheet absorbs ≈ 7 × more energy per unit area.

4A. Applications of the Westergaard Method

Different analytic $Z_I(z)$ give K for different **Mode I** loadings of a crack in an infinite plate:

Loading on crack	Westergaard function / result
Remote biaxial σ	$Z = \sigma z/\sqrt{(z^2-a^2)} \Rightarrow K = \sigma\sqrt{(\pi a)}$
Uniform pressure p on crack faces	same as remote $\sigma = p \Rightarrow$ **$K = p\sqrt{(\pi a)}$**
Pair of splitting forces P (per unit thickness) at x = b	$K_{-\pm} = (P/\sqrt{(\pi a)})\ \sqrt{[(a \pm b)/(a \mp b)]}$; at centre (b = 0): **$K = P/\sqrt{(\pi a)}$**
Periodic array of collinear cracks (2a, pitch 2W)	$K = \sigma\sqrt{(\pi a)} \cdot \sqrt{[(2W/\pi a)\tan(\pi a/2W)]}$
Finite width strip (Feddersen / secant)	$K = \sigma\sqrt{(\pi a)}\ \sqrt{(\sec(\pi a/W))}$ (W = width, 2a crack)

- For P at the **centre** the point load is carried to the tips; K falls as the crack grows (the

wedge-force effect), so growth is stable in wedge-loaded specimens – used in DCB / wedge-opening-load (WOL) tests.

- Collinear array: interaction between neighbouring tips raises K above $\sigma\sqrt{(\pi a)}$ (factor → 1 as a/W → 0 and → ∞ as 2a → pitch).
- **Limitations of the original Westergaard method:** (i) only symmetric (Mode I) problems with $\sigma_x = \sigma_y$ at infinity; for $\sigma_x \neq \sigma_y$ one adds a constant (Eftis–Liebowitz): $\sigma_x = \text{Re } Z - y \text{ Im } Z' - A$, where A = difference of remote stresses — the **T-stress** term; (ii) boundary conditions satisfied only at the crack faces and infinity – finite geometry corrections need other methods.

Mode II and III (preview of Unit IV)

- Mode II: analogous complex function Z_{II} gives $K_{II} = \tau\sqrt{(\pi a)}$ and $\sigma_{ij} = K_{II}/\sqrt{(2\pi r)}\ f_{ij}^{II}(\theta)$.
- Mode III: displacement w(x, y) is harmonic ⇒ τ_{xz}, τ_{yz} from an analytic function; $K_{III} = \tau\sqrt{(\pi a)}$.

4B. How is K Determined?

Method	Principle
Analytical (Westergaard, Muskhelishvili)	Closed form for simple geometries (infinite plate); handbooks (Tada, Rooke, Murakami)
Compliance method	Measure C(a); $G = (P^2/2B)\ dC/da$; $K = \sqrt{(E'G)}$
Displacement extrapolation (FEM / DIC)	$K = [2\mu/(\kappa+1)]\ \sqrt{(2\pi/r)} \cdot v(r, \theta=\pi)$, $r \to 0$
Stress extrapolation	$K = \lim \sigma_y \sqrt{(2\pi r)}$ as $r \to 0$, $\theta = 0$
Virtual crack closure (VCCT)	$G_I \approx F_y \Delta v / (2B\Delta a)$ from nodal force & opening
J-integral (domain integral)	$J = G \Rightarrow K$; path independent – most accurate in FEM
Photoelasticity / caustics / strain gauges	Isochromatic fringes, shadow-spot (caustic) diameter, gauges at known (r, θ) near tip

- FEM uses **quarter-point (Barsoum / Henshell–Shaw) singular elements**: mid-side nodes shifted to ¼ of the side from the tip give a $1/\sqrt{r}$ strain singularity.
- Check with Example 2: $K = 2\mu\ v/(\kappa+1) \cdot \sqrt{(2\pi/r)} = 2\times26.3\times10^{9}\times1.93\times10^{-5}/2.68 \times \sqrt{(2\pi/10^{-3})} \approx 30$ MPa√m ✔.

4C. More on Elasticity Basics

A. Stress transformation & Mohr's circle

$$\sigma_x' = (\sigma_x+\sigma_y)/2 + (\sigma_x-\sigma_y)/2 \cdot \cos 2\alpha + \tau_{xy} \sin 2\alpha$$

$$\tau_{x'y'} = -(\sigma_x-\sigma_y)/2 \cdot \sin 2\alpha + \tau_{xy} \cos 2\alpha \; ; \; \tan 2\alpha_p = 2\tau_{xy}/(\sigma_x-\sigma_y)$$

- Mohr's circle: centre at $(\sigma_x+\sigma_y)/2$, radius = τ_{max}. At the crack line $\theta = 0$, $\sigma_1 = \sigma_2 = K/\sqrt{(2\pi r)}$: the circle degenerates into a **point** (equi-biaxial). In plane strain $\sigma_3 = 2\nu\sigma_1$ so the max shear stress is small ⇒ yielding is suppressed ⇒ brittle behaviour.

B. Elastic constants & energy

- E, ν, $\mu = E/2(1+\nu)$, bulk modulus $K_b = E/3(1-2\nu)$, Lamé constant $\lambda = E\nu/[(1+\nu)(1-2\nu)]$. Only two are independent for an isotropic solid.
- Strain-energy density $W = \tfrac{1}{2}\, \sigma_{ij}\, \varepsilon_{ij}$; plane stress: $W = (1/2E)[\sigma_x^2 + \sigma_y^2 - 2\nu\sigma_x\sigma_y + 2(1+\nu)\tau_{xy}^2]$. Near the tip $W \propto 1/r$.
- **Saint-Venant's principle:** effects of load details are local; far-field stress depends only on resultants. **Superposition:** linear elasticity lets us add solutions ($K_{total} = K_a + K_b$ for the same mode).
- Boundary-value problems: traction BC, displacement BC, mixed. Crack faces: $\sigma_y = \tau_{xy} = 0$ (traction-free) for $|x| < a$.

C. Why the inverse-square-root singularity?

Any solution of $\nabla^4\varphi = 0$ with traction-free flanks ($\theta = \pm\pi$) has eigen-expansion $\varphi = r^{\lambda+1} F(\theta)$. Finite displacement/energy at the tip needs $\mathrm{Re}\,\lambda > 0$, and the **lowest admissible** exponent is $\lambda = 1/2 \Rightarrow \sigma \sim r^{\lambda-1} = r^{-1/2}$ (Williams, 1957). Higher terms (r^0, $r^{1/2}$, …) give the T-stress and multi-parameter fields (Unit IV).

4D. Extra Solved Problems

P1. Plane stress, $K_I = 50$ MPa$\sqrt{m}$, $\sigma_{ys} = 400$ MPa. At $r = 0.5$ mm, $\theta = 0$: $\sigma_y = 50/\sqrt{(2\pi\times0.0005)} = $ **892 MPa** $> \sigma_{ys}$ ⇒ elastic solution invalid there; Irwin plastic radius $r_y = (1/2\pi)(K/\sigma_{ys})^2 = (1/2\pi)(0.125)^2 = $ **2.5 mm**; the elastic K-field is valid only well outside r_y.

P2 (strain-gauge K). Measured $\sigma_y = 250$ MPa at $r = 3$ mm, $\theta = 0$ (K-dominant region): $K_I = \sigma_y\sqrt{(2\pi r)} = 250\times\sqrt{(2\pi\times0.003)} = 250\times0.1373 = $ **34.3 MPa$\sqrt{m}$**.

P3 (G_c). Aluminium sheet $G_c = 24$ kJ/m², $E = 72$ GPa, half-crack $a = 15$ mm. $\sigma_f = \sqrt{(E\, G_c/(\pi\, a))} = \sqrt{(72\times10^9\times24000/(\pi\times0.015))} = $ **191 MPa**.

P4 (periodic cracks). Cracks $2a = 20$ mm, pitch $2W = 100$ mm, $\sigma = 100$ MPa. $Y = \sqrt{[(2W/\pi a)\tan(\pi a/2W)]} = \sqrt{[(100/31.42)\tan(0.3142)]} = \sqrt{(3.183\times0.3249)} = 1.017 \Rightarrow K =$

$100\sqrt{(\pi\times0.01)}\times1.017 = 17.72\times1.017 =$ **18.0 MPa√m**.

P5 (splitting force). $P = 5\times10^6$ N/m (per unit thickness) at the centre, $a = 25$ mm: $K = P/\sqrt{(\pi a)} = 5\times10^6/\sqrt{(\pi\times0.025)} =$ **17.8 MPa√m**.

P6. A $2a = 8$ mm crack, $\sigma = 120$ MPa, $K_{Ic} = 45$ MPa√m. $K = 120\sqrt{(\pi\times0.004)} = 13.5$ MPa√m ⇒ safety factor on $K = 45/13.5 =$ **3.3**; $a_c = (1/\pi)(45/120)^2 =$ **44.8 mm**.

4E. Model Answer – Westergaard Derivation in 8 Steps

1. State assumptions: infinite plate, 2a crack on x-axis, biaxial σ, Mode I, plane stress/strain.
2. Airy function $\varphi = \mathrm{Re}\, Z_2 + y\, \mathrm{Im}\, Z_1$ with $Z_1' = Z$, $Z_2' = Z_1 \Rightarrow \nabla^4\varphi = 0$ automatically.
3. Stresses: $\sigma_x = \mathrm{Re}\, Z - y\, \mathrm{Im}\, Z'$, $\sigma_y = \mathrm{Re}\, Z + y\, \mathrm{Im}\, Z'$, $\tau_{xy} = -y\, \mathrm{Re}\, Z'$.
4. Choose $Z = \sigma z/\sqrt{(z^2-a^2)}$; verify B.C.: (i) $|z| \to \infty$ gives σ; (ii) crack faces traction-free.
5. Shift origin to the tip: $z = a + \zeta$, $Z \approx \sigma\sqrt{(a/2\zeta)} = K/\sqrt{(2\pi\zeta)}$, $K = \sigma\sqrt{(\pi a)}$.
6. Write $\zeta = r\, e^{i\theta}$, separate real & imaginary parts ⇒ σ_x, σ_y, τ_{xy} (boxed results).
7. Integrate Z for displacements u, v (with κ).
8. Conclude: $\sigma_{ij} = K/\sqrt{(2\pi r)}\, f_{ij}(\theta)$; K = single parameter; $G = K^2/E'$.

Quick MCQs

1) Airy stress function satisfies : **$\nabla^4\varphi = 0$**. 2) Westergaard's Z_1 is analytic **except at the crack tips** (branch points $z = \pm a$).

3) For plane strain $E' =$ **$E/(1-\nu^2)$**. 4) Order of singularity of σ at a crack tip : **$r^{-1/2}$**. 5) K–G relation : **$G = K^2/E'$**.

4F. Angular Variation of the K-field

Let $A = K_I/\sqrt{(2\pi r)}$. Values of the angular functions (plane stress/strain, Mode I):

θ (deg)	σ_x / A	σ_y / A	τ_{xy} / A
0	1.000	1.000	0.000
30	0.789	1.143	0.177
60	0.433	1.299	0.000
90	0.354	1.061	-0.354
120	0.500	0.500	-0.433
150	0.436	0.082	-0.177
180	0.000	0.000	0.000

- σ_y is largest on the crack line (θ = 0) and falls to 0 on the faces (θ = 180°) – free

surface. $\sigma_x \to 0$ at 180° too.

- **Max shear stress** occurs at $\theta \approx \pm 70°$ (plane stress); in plane strain it is much smaller because $\sigma_3 \approx 2\nu A$ is close to σ_1 – this is why plane-strain plastic zones are small and toughness is low.
- The **tangential (hoop) stress** for pure Mode I is $\sigma_{\theta\theta} = (K_I/\sqrt{(2\pi r)})\cos^3(\theta/2)$ (used for the mixed-mode kink angle, Unit V); its maximum is at $\theta = 0 \Rightarrow$ the crack grows straight ahead (self-similar).

A. Derivation of displacements (sketch)

1. Strains from Hooke's law (plane stress): $\varepsilon_x = (\sigma_x - \nu\sigma_y)/E$, $\varepsilon_y = (\sigma_y - \nu\sigma_x)/E$.
2. Integrate $\varepsilon_x = \partial u/\partial x$, $\varepsilon_y = \partial v/\partial y$ using the Westergaard stresses.
3. Determine rigid-body terms by requiring symmetry: $v(x, 0) = 0$ ahead of the tip and $u(0, y) = 0$ reference.
4. Result in compact form using Z_1: $2\mu u = ((\kappa-1)/2)\,\mathrm{Re}\,Z_1 - y\,\mathrm{Im}\,Z$; $2\mu v = ((\kappa+1)/2)\,\mathrm{Im}\,Z_1 - y\,\mathrm{Re}\,Z$.
5. Insert $Z_1 = K\sqrt{(2\zeta/\pi)}$, $\zeta = r\,e^{i\theta} \Rightarrow u, v$ given in Section 3.

B. Westergaard vs Williams vs Irwin approach

Approach	Idea	Result / use
Westergaard (1939)	Single analytic function $Z(z)$ for symmetric problems	Closed form K for infinite-plate cases
Irwin (1957)	Used Westergaard near tip, defined K; $G = K^2/E'$	Founded LEFM; K-based design
Williams (1957)	Eigenfunction series $r^{n/2}$ for wedge with free flanks	Gives K, T-stress and higher-order terms
Muskhelishvili	Two complex potentials (general plane problem)	All 3 modes, finite bodies, holes

C. Practice Questions with Answers

1. At $r = 1$ mm directly ahead of a crack with $K_I = 25$ MPa$\sqrt{m}$, find σ_y. Ans: $25/\sqrt{(2\pi\times0.001)} =$ **315 MPa**.
2. Find the critical stress for a $2a = 30$ mm crack in glass ($E = 70$ GPa, $G_c = 8$ J/m²), plane stress. Ans: $\sqrt{(E\,G_c/(\pi a))} = \sqrt{(70\times10^9\times8/(\pi\times0.015))} =$ **3.45 MPa**.
3. Find G and K for $\sigma = 80$ MPa, $2a = 60$ mm, $E = 200$ GPa, $\nu = 0.3$ (plane strain). Ans: $K = 80\sqrt{(\pi\times0.03)} =$ **24.6 MPa$\sqrt{m}$**; $G = (1-\nu^2)K^2/E = 0.91\times(24.6\times10^6)^2/(200\times10^9) =$ **2.74 kJ/m²**.
4. If K_I doubles, what happens to the crack-tip stress at fixed r and to G? Ans: stress

doubles; G quadruples ($\propto K^2$).

5. For $K_I = 30$ MPa$\sqrt{m}$ ($E = 70$ GPa, plane stress) what is the CTOD at $r = 0.5$ mm behind the tip? Use $\delta = 2v = (\kappa+1)K/\mu\ \sqrt{(r/2\pi)}$ with $\kappa+1 = 4/(1+\nu)$. Ans ($\nu = 0.33$): $\mu = 26.3$ GPa, $\kappa+1 = 3.01 \Rightarrow \delta = 3.01\times30\times10^6/(26.3\times10^9) \times \sqrt{(0.0005/2\pi)} = 3.43\times10^{-3} \times 0.00892 = \mathbf{3.1\times10^{-5}\ m \approx 31\ \mu m}$.

5. Summary & Exam Pointers

Quantity	Expression (Mode I)
Westergaard function	$Z_I = \sigma z/\sqrt{(z^2-a^2)}$ (centre crack, biaxial)
Stresses	$\sigma_x = \mathrm{Re}\ Z - y\ \mathrm{Im}\ Z'$; $\sigma_y = \mathrm{Re}\ Z + y\ \mathrm{Im}\ Z'$; $\tau_{xy} = -y\ \mathrm{Re}\ Z'$
Tip field	$\sigma_{ij} = K/\sqrt{(2\pi r)}\ f_{ij}(\theta)$, $K = \sigma\sqrt{(\pi a)}$
Displacements	$\propto K\sqrt{(r/2\pi)}\ g_{ij}(\theta)/(2\mu)$, $\kappa = 3-4\nu$ or $(3-\nu)/(1+\nu)$
Crack opening	$v = (2\sigma/E')\sqrt{(a^2-x^2)}$
K–G	$G = K^2/E'$

1. Derive the stress field near a crack tip for Mode I using the Westergaard stress function.
2. Explain the Airy stress function. Why does $\nabla^4\varphi = 0$?
3. Derive $G = K^2/E'$ (plane stress) using Irwin's crack-closure integral.
4. State the boundary conditions satisfied by Westergaard's function for a centre crack in an infinite plate.
5. Explain plane stress / plane strain and the corresponding values of E', κ.
6. Sketch crack-opening displacement and show it is elliptical.

UNIT 4

Multi-parameter Stress Field, Mixed Modes, SIFs & Plastic Zone Models

Williams series | T-stress | Modes II & III | mixed mode | SIF handbook | Irwin | Dugdale

1. Multi-parameter Stress Field (Mode I)

The Westergaard/Irwin field $\sigma_{ij} = K/\sqrt{(2\pi r)}\ f_{ij}(\theta)$ is only the **first term** of an infinite series. Near the tip ($r \to 0$) it dominates, but at finite r (small specimens, short cracks, experiments) higher-order terms matter.

1.1 Williams eigenfunction expansion (1957)

Solve $\nabla^4\varphi = 0$ for a wedge of 2π with traction-free flanks, $\varphi = r^{n/2+1}\ F_n(\theta)$. For Mode I (symmetric):

$$\sigma_x = \Sigma_n\ A_n\ (n/2)\ r^{n/2-1}\ \{\ [2 + n/2 + (-1)^n]\ \cos(n/2-1)\theta - (n/2-1)\ \cos(n/2-3)\theta\ \}$$

$$\sigma_y = \Sigma_n\ A_n\ (n/2)\ r^{n/2-1}\ \{\ [2 - n/2 - (-1)^n]\ \cos(n/2-1)\theta + (n/2-1)\ \cos(n/2-3)\theta\ \}$$

$$\tau_{xy} = \Sigma_n\ A_n\ (n/2)\ r^{n/2-1}\ \{\ [n/2 + (-1)^n]\ \sin(n/2-1)\theta - (n/2-1)\ \sin(n/2-3)\theta\ \}$$

Term n	Order in r	Meaning
1	$r^{-1/2}$	singular term: $A_1 = K_I/\sqrt{(2\pi)}$ → K-field
2	r^0	uniform stress parallel to crack: $\sigma_{x0} = T = 4A_2$ (**T-stress**)
3	$r^{1/2}$	first non-singular higher-order term, A_3 …
≥ 4	r^1, $r^{3/2}$ …	further terms; matter far from tip

Why multi-parameter fields?

- Single-term (K) data fit only for $r \to 0$, but near the tip plasticity, 3-D effects and measurement noise spoil data; far data need more terms.
- Photoelastic / strain-gauge / DIC measurements are taken at r = 0.3–1 × thickness: using 3–5 terms with over-deterministic **least-squares** (Sanford & Dally; Ramesh) gives accurate K and T.
- Short cracks, edge cracks and specimen geometry change the higher terms ⇒ explain size/geometry (**constraint**) effects on toughness.

1.2 T-stress and constraint

Three-term field: $\sigma_{ij} = K_I/\sqrt{(2\pi r)}\, f_{ij}(\theta) + T\,\delta_{1i}\delta_{1j} + O(r^{1/2})$. T acts parallel to the crack plane (σ_x).

$$\text{Biaxiality ratio } B = T\sqrt{(\pi a)} / K_I$$

- Centre crack, uniaxial $\sigma \perp$ crack: $T = -\sigma$ ($B = -1$). Equi-biaxial loading: $T = 0$. Edge-cracked bend (SENB) deep crack: $T \approx -0.4\sigma$... (negative, **low constraint**). Deep CT: T slightly positive (high constraint).
- **$T > 0$** (high constraint): plastic zone elongates forward, high triaxiality, lower toughness; **$T < 0$** (low constraint): zone rotates/enlarges, triaxiality lost, toughness higher.
- T-stress also governs **directional stability** of crack path: $T > 0$ → path unstable (deviates), $T < 0$ → stable.
- Two-parameter approaches: **K–T** (elastic) and **J–Q** (O'Dowd & Shih; Q = difference from SSY field). They explain why small specimens (low constraint) give higher apparent toughness than structures.

2. Mode II, Mode III and Mixed Modes

2.1 Mode II (in-plane shear) near-tip field

$$\sigma_x = -K_{II}/\sqrt{(2\pi r)} \cdot \sin(\theta/2)\,[\,2 + \cos(\theta/2)\cos(3\theta/2)\,]$$

$$\sigma_y = K_{II}/\sqrt{(2\pi r)} \cdot \sin(\theta/2)\cos(\theta/2)\cos(3\theta/2)$$

$$\tau_{xy} = K_{II}/\sqrt{(2\pi r)} \cdot \cos(\theta/2)\,[\,1 - \sin(\theta/2)\sin(3\theta/2)\,]$$

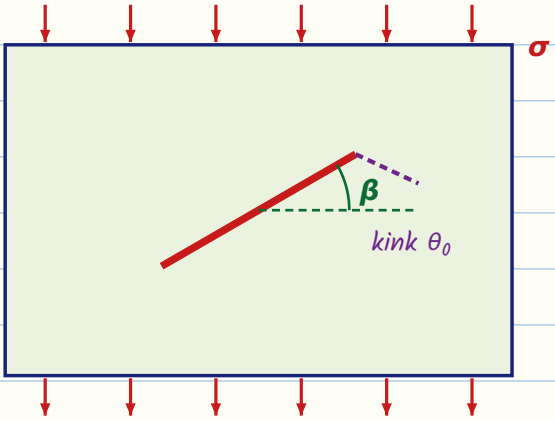
- $K_{II} = \tau\sqrt{(\pi a)}$ (centre crack, remote shear τ). Displacements: $u = (K_{II}/2\mu)\sqrt{(r/2\pi)}\,\sin(\theta/2)[\kappa + 1 + 2\cos^2(\theta/2)]$, $v = -(K_{II}/2\mu)\sqrt{(r/2\pi)}\,\cos(\theta/2)[\kappa - 1 - 2\sin^2(\theta/2)]$.
- On crack line $\theta = 0$: $\sigma_x = \sigma_y = 0$ and $\tau_{xy} = K_{II}/\sqrt{(2\pi r)}$ — pure shear ahead of tip. σ_x is antisymmetric ⇒ maximum tension is **off-axis**.

2.2 Mode III (anti-plane shear)

$$\tau_{xz} = -K_{III}/\sqrt{(2\pi r)}\,\sin(\theta/2)\ ;\ \tau_{yz} = K_{III}/\sqrt{(2\pi r)}\,\cos(\theta/2)\ ;\ w = (2K_{III}/\mu)\sqrt{(r/2\pi)}\,\sin(\theta/2)$$

- $K_{III} = \tau\sqrt{(\pi a)}$. Only one displacement $w(x, y)$ ⇒ simplest: governed by Laplace equation $\nabla^2 w = 0$. $G_{III} = K_{III}^2/(2\mu) = (1+\nu)K_{III}^2/E$.

2.3 Mixed-mode loading



Inclined crack (β from ⊥ to σ):

$\sigma_n = \sigma \cos^2\beta$, $\tau = \sigma \sin\beta \cos\beta$

$K_I = \sigma\sqrt{(\pi a)}\cos^2\beta$

$K_{II} = \sigma\sqrt{(\pi a)}\sin\beta\cos\beta$

$\beta = 0$: pure Mode I
$\beta = 45°$: $K_I = 0.5K$, $K_{II} = 0.5K$
$\beta = 90°$: crack ∥ load, $K = 0$

Fig 4.1 Inclined crack – mixed Mode I + II

- Superposition: $\sigma_{ij} = [K_I f^{\wedge}I_{ij} + K_{II} f^{\wedge}II_{ij} + K_{III} f^{\wedge}III_{ij}]/\sqrt{(2\pi r)}$. Energy: $G = (K_I^2 + K_{II}^2)/E' + K_{III}^2/2\mu$.
- Mode-mixity angle $\psi = \tan^{-1}(K_{II}/K_I)$. Real structures are rarely pure Mode I (inclined cracks, welds, shafts under torsion + bending).
- Under mixed I + II the crack **kinks** and grows at an angle θ_0 to its original plane; the criteria (MTS, SED, G_{max}) are discussed in Unit V.

Worked Example 1 (inclined crack)

Central crack $2a = 20$ mm inclined $\beta = 30°$ to the plane ⊥ load, $\sigma = 100$ MPa (infinite plate, $E' = 210$ GPa). $K_0 = \sigma\sqrt{(\pi a)} = 100\sqrt{(\pi \times 0.01)} = 17.72$ MPa√m.

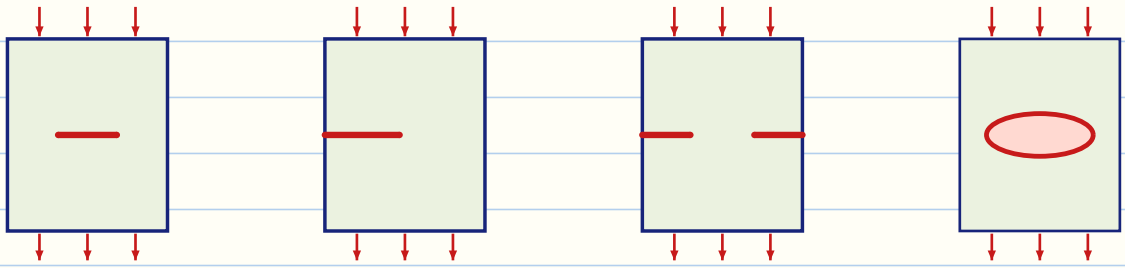
$K_I = K_0 \cos^2 30° =$ **13.29 MPa√m** ; $K_{II} = K_0 \sin 30° \cos 30° =$ **7.67 MPa√m** ; $G = (K_I^2 + K_{II}^2)/E' = (13.29^2 + 7.67^2)\times10^{12}/(210\times10^9) =$ **1.12 kJ/m²**.

MTS kink angle: $\theta_0 = 2\tan^{-1}[(K_I - \sqrt{(K_I^2+8K_{II}^2)})/(4K_{II})] = 2\tan^{-1}[(13.29 - 25.44)/30.70]$ = **−43.2°** (crack turns about 43° from its plane).

3. Stress Intensity Factors for Various Geometries

$$K_I = Y \sigma \sqrt{(\pi a)}$$

Y = geometry (correction) factor



Centre crack (CCT)	Single edge (SENT)	Double edge (DENT)	Embedded penny
$Y = \sqrt{\sec(\pi a/W)}$	$Y \approx 1.12$ $(a/W \to 0)$	$Y \approx 1.12$ $(a/W \to 0)$	$K = 2\sigma\sqrt{(a/\pi)}$

Fig 4.2 Common cracked geometries

Geometry	Y or K expression
Centre crack 2a, finite width W (Feddersen)	$Y = \sqrt{\sec(\pi a/W)}$ $(a/W < \sim 0.7)$
Single edge crack (SENT, $a/W \le 0.6$)	$Y = 1.12 - 0.231(a/W) + 10.55(a/W)^2 - 21.72(a/W)^3 + 30.39(a/W)^4$; small a: **1.12** (free-surface factor)
Double edge cracks, small a/W	$Y \approx 1.12$
Embedded penny-shaped crack, radius a	$K = 2\sigma\sqrt{(a/\pi)} = (2/\pi)\,\sigma\sqrt{(\pi a)} = 0.637\,\sigma\sqrt{(\pi a)}$
Embedded elliptical crack ($a < c$)	$K = \sigma\sqrt{(\pi a/Q)}\,[\sin^2\varphi + (a^2/c^2)\cos^2\varphi]^{1/4}$, $Q \approx 1 + 1.464\,(a/c)^{1.65}$
Semi-elliptical surface crack (deepest point)	$K \approx 1.12\,\sigma\,\sqrt{(\pi a/Q)}$
Thin cylinder, long. through crack (half-length a)	$K = (pR/t)\,\sqrt{(\pi a)} \cdot M$, $M \approx \sqrt{(1 + 1.61\,a^2/(Rt))}$ (Folias bulging)
Cracks at a hole (short, symmetric)	$K \approx 3.39\,\sigma\,\sqrt{(\pi a)}$ $(a \ll r)$; for $a \gg r$ treat as crack $2(a + r)$
Pressure p in crack / wedge force P at centre	$K = p\sqrt{(\pi a)}$; $K = P/\sqrt{(\pi a)}$ (P per unit thickness)

3.1 Standard test specimens (ASTM E399) – $\alpha = a/W$

$$CT: K_I = \frac{P}{B\sqrt{W}} \cdot f(\alpha)$$

$f(\alpha) = (2+\alpha)(0.886 + 4.64\alpha - 13.32\alpha^2 + 14.72\alpha^3 - 5.6\alpha^4) \,/\, (1-\alpha)^{3/2}$

$$SENB\ (3\text{-pt},\ S = 4W): K_I = \frac{P\,S}{B\,W^{3/2}} \cdot f(\alpha)$$

$f(\alpha) = 3\sqrt{\alpha}\,[1.99 - \alpha(1-\alpha)(2.15 - 3.93\alpha + 2.7\alpha^2)] \,/\, [2(1+2\alpha)(1-\alpha)^{3/2}]$

- At $\alpha = 0.5$: $f_{CT} = 9.66$, $f_{SENB} = 2.66$.
- **Superposition:** K for combined loads = ΣK_i (same mode). Residual stress, thermal stress and pressure can each be handled separately. **Weight functions** give K for any stress distribution along crack line.
- Sources: Tada–Paris–Irwin handbook, Rooke & Cartwright, Murakami, Sih; FE (quarter-point elements, J-integral).

Worked Example 2 (SENT)

Strip $W = 50$ mm, edge crack $a = 10$ mm ($a/W = 0.2$), $\sigma = 200$ MPa. $Y = 1.12 - 0.0462 + 0.4220 - 0.1738 + 0.0486 = 1.371 \Rightarrow K = 1.371 \times 200 \times \sqrt{(\pi \times 0.01)} =$ **48.6 MPa√m**.

Worked Example 3 (CCT)

Plate $W = 100$ mm, $2a = 40$ mm, $\sigma = 150$ MPa. $\pi a/W = \pi \times 0.02/0.1 = 0.628$ rad $\Rightarrow \sec = 1.236$, $Y = \sqrt{1.236} = 1.112 \Rightarrow K = 1.112 \times 150 \times \sqrt{(\pi \times 0.02)} =$ **41.8 MPa√m**. (Infinite plate would give 37.6: finite width raises K by 11 %.)

Worked Example 4 (CT specimen)

$W = 50$ mm, $B = 25$ mm, $a = 25$ mm ($\alpha = 0.5$), $P = 20$ kN. $P/(B\sqrt{W}) = 20000/(0.025 \times 0.2236) = 3.58 \times 10^6$ Pa√m $\Rightarrow K = 3.58 \times 10^6 \times 9.66 =$ **34.6 MPa√m**.

Worked Example 5 (pressure vessel)

Thin cylinder $R = 500$ mm, $t = 10$ mm, $p = 5$ MPa $\Rightarrow \sigma_h = pR/t = 250$ MPa. Longitudinal through crack, half-length $a = 15$ mm: $M = \sqrt{(1 + 1.61 \times 0.015^2/(0.5 \times 0.01))} = 1.036 \Rightarrow K = 250\sqrt{(\pi \times 0.015)} \times 1.036 =$ **56.2 MPa√m**. If $K_{Ic} = 60$ MPa√m, margin is slim ⇒ **leak-before-break** check needed.

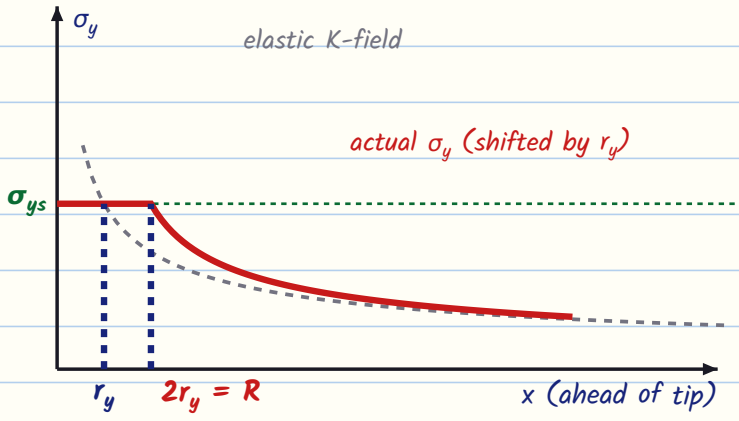
Worked Example 6 (surface crack)

Semi-elliptical surface crack $a = 5$ mm, $2c = 20$ mm ($a/c = 0.5$), $\sigma = 150$ MPa. $Q = 1 + 1.464(0.5)^{1.65} = 1.466 \Rightarrow K = 1.12 \times 150 \times \sqrt{(\pi \times 0.005/1.466)} =$ **17.4 MPa√m** at the deepest point.

4. Modelling Plastic Deformation at the Crack Tip

Elastic solution gives $\sigma_y \to \infty$ at tip, but real metals yield at $\sigma_{ys} \Rightarrow$ a **plastic zone** forms. If its size $r_p \ll a$ (small-scale yielding) LEFM remains valid with a small correction.

4.1 Irwin's model (1960)



Irwin's argument

$\sigma_y = K/\sqrt{(2\pi r)} = \sigma_{ys}$ at $r = r_y$

Plane stress:

$$r_y = (1/2\pi)(K/\sigma_{ys})^2$$

Load carried by cut-off area is redistributed ⇒ true zone = $2r_y$

Effective crack: $a_{eff} = a + r_y$

Fig 4.3 Irwin's plastic-zone correction

1. Elastic stress on crack line $\sigma_y = K/\sqrt{(2\pi r)}$. Yielding when $\sigma_y = \sigma_{ys} \Rightarrow r = r_y = (1/2\pi)(K/\sigma_{ys})^2$ (first estimate; plane stress).

2. In the yielded region σ_y cannot exceed σ_{ys}; the excess load (area under elastic curve above σ_{ys}) must be carried farther ahead ⇒ the zone doubles: **plastic zone size $2r_y = (1/\pi)(K/\sigma_{ys})^2$** (the equilibrium argument: $\sigma_{ys}\cdot R = \int_0^{r-y} K/\sqrt{(2\pi r)}\,dr \Rightarrow R = 2r_y$).
3. The plastic zone makes the crack behave as if it were **longer**: effective crack length **$a_{eff} = a + r_y$** (tip moves to the middle of the zone).

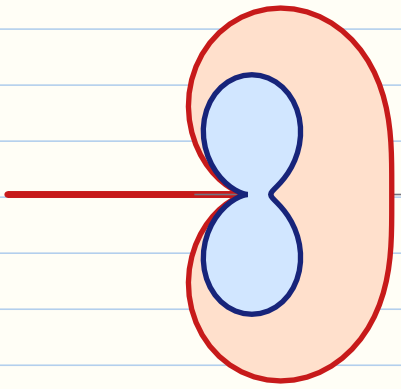
$$K_{eff} = Y\,\sigma\,\sqrt{(\pi\,(a + r_y))}$$

iterate; for CCT: $K_{eff} = K/\sqrt{(1 - \tfrac{1}{2}(\sigma/\sigma_{ys})^2)}$

Irwin plastic-zone formulae

> Plane stress: $r_y = (1/2\pi)(K/\sigma_{ys})^2$; zone size $2r_y = (1/\pi)(K/\sigma_{ys})^2$.
> Plane strain (σ_y constrained to ≈ $\sqrt{3}\,\sigma_{ys}$): $r_y = (1/6\pi)(K/\sigma_{ys})^2$ ≈ 1/3 of plane stress.
> CTOD (Irwin): $\delta_t = 2v(r_y) = (4/\pi)\,K^2/(E\,\sigma_{ys})$.
> Thickness: zone ≈ B ⇒ plane stress; $B \geq 2.5(K_{Ic}/\sigma_{ys})^2$ ⇒ plane strain.

4.2 Shape of the plastic zone



plane stress (large, 'dog-bone'/ round)

plane strain (small, 'butterfly' lobes)

von Mises, $S = K^2/4\pi\sigma_{ys}^2$

σ-state: $r_p(\theta) = S[1 + 1.5\sin^2\theta + \cos\theta]$

ε-state: $r_p(\theta) = S[1.5\sin^2\theta + (1-2\nu)^2(1+\cos\theta$

Fig 4.4 von Mises plastic zones – plane stress vs plane strain

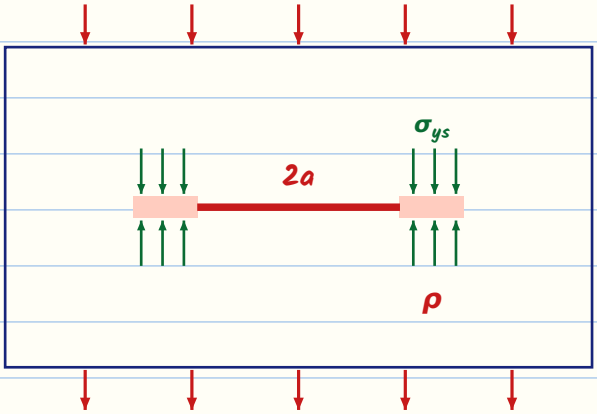
- Tresca & von Mises criteria applied to the K-field give zone shapes: plane stress – large, extends mostly **ahead & at ≈ 70°**; plane strain – small, **butterfly** lobes tilted back at 70°, almost no extent ahead ($\sigma_3 = 2\nu\sigma$ reduces shear).
- Hardening materials: smaller zone; large zone at high σ/σ_{ys} (>0.5) ⇒ need EPFM.
- Cyclic loading: reversed (cyclic) plastic zone $r_c \approx (1/2\pi)(\Delta K/2\sigma_{ys})^2$ = **¼ of the monotonic zone at ΔK**; controls closure & retardation (Unit V).

Worked Example 7

Plate, σ = 200 MPa, $2a$ = 40 mm, σ_{ys} = 400 MPa. $K = 200\sqrt{(\pi\times0.02)} = 50.1$ MPa√m.

r_y(stress) $= (1/2\pi)(50.1/400)^2 =$ **2.5 mm** (plane strain 0.83 mm). $a_{eff} = 22.5$ mm ⇒ K_{eff} ≈ **53.6 MPa√m** (7 % higher); CTOD (Irwin) $= (4/\pi)K^2/(E\sigma_{ys}) =$ **0.038 mm**.

4.3 Dugdale strip-yield model (1960)



Dugdale (strip-yield) model:

thin plastic strip ρ at each tip,
closing stress = σ_{ys} (perfectly plastic)

Real crack 2a → fictitious 2(a+ρ)

No stress singularity: $K_\sigma + K_{closure} = 0$

$\rho = a\,[\sec(\pi\sigma/2\sigma_{ys}) - 1]$

$\delta = (8\sigma_{ys}\,a/\pi E)\ln\sec(\pi\sigma/2\sigma_{ys})$

Fig 4.5 Dugdale model – thin plane-stress sheets

Assumptions: thin sheet (plane stress), **elastic-perfectly plastic**, plasticity confined to a thin strip of length ρ ahead of each tip, strip carries uniform stress σ_{ys}, crack behaves as one of length $2c = 2(a + \rho)$.

1. Problem (a): crack 2c under remote $\sigma \Rightarrow K_\sigma = \sigma\sqrt{(\pi c)}$.
2. Problem (b): crack 2c with closing stress σ_{ys} on the strips $a < |x| < c \Rightarrow K_{closure} = -2\sigma_{ys}\sqrt{(c/\pi)}\cos^{-1}(a/c)$.
3. Stress finite at the tip $\Rightarrow K_\sigma + K_{closure} = 0 \Rightarrow \sigma\sqrt{(\pi c)} = 2\sigma_{ys}\sqrt{(c/\pi)}\cos^{-1}(a/c)$.

$$a/c = \cos(\pi\sigma / 2\sigma_{ys}) \Rightarrow \rho = a\,[\sec(\pi\sigma/2\sigma_{ys}) - 1]$$

$$\delta_t = (8\sigma_{ys}\,a/\pi E)\ln\sec(\pi\sigma/2\sigma_{ys}) \quad \text{CTOD}$$

- **Small σ/σ_{ys}:** $\rho \approx (\pi^2/8)\,a\,(\sigma/\sigma_{ys})^2 = (\pi/8)(K/\sigma_{ys})^2$; $\delta_t \approx K^2/(E\,\sigma_{ys}) = G/\sigma_{ys}$.
- As $\sigma \to \sigma_{ys}$: $\rho \to \infty$ and $\delta \to \infty$ (plastic collapse). Critical stress from CTOD criterion $\delta = \delta_c$: $\sigma_c = (2\sigma_{ys}/\pi)\cos^{-1}[\exp(-\pi E\delta_c/8\sigma_{ys}\,a)]$.

Aspect	Irwin	Dugdale
Zone ($\sigma << \sigma_{ys}$)	$2r_y = K^2/\pi\sigma_{ys}^2 = 0.318\,K^2/\sigma_{ys}^2$	$\rho = (\pi/8)K^2/\sigma_{ys}^2 = 0.393\,K^2/\sigma_{ys}^2$
CTOD ($\sigma << \sigma_{ys}$)	$(4/\pi)\,K^2/E\sigma_{ys} = 1.27\,K^2/E\sigma_{ys}$	$K^2/(E\sigma_{ys})$
Material	elastic + yield cut-off	elastic-perfectly plastic strip
Valid range	$\sigma/\sigma_{ys} < \sim 0.5$	all $\sigma/\sigma_{ys} < 1$ (thin sheets)

Worked Example 8 (Dugdale)

$\sigma/\sigma_{ys} = 0.5 \Rightarrow \rho = a(\sec 45° - 1) = \mathbf{0.414\,a}$; $\sigma/\sigma_{ys} = 0.8 \Rightarrow \rho = a(\sec 72° - 1) = \mathbf{2.24\,a}$ (zone larger than the crack!).

Critical stress: E = 210 GPa, σ_{ys} = 400 MPa, a = 25 mm, δ_c = 0.1 mm: exponent = $\pi\times210\times10^9\times10^{-4}/(8\times400\times10^6\times0.025) = 0.825 \Rightarrow \exp(-0.825) = 0.438 \Rightarrow \cos^{-1} = 1.117$ rad ⇒

$\sigma_c = (2\times400/\pi)\times1.117 =$ **284 MPa** ($0.71\ \sigma_{ys}$).

4A. More on Multi-parameter Fields and SIFs

A. How the Williams series arises (outline)

1. Take $\varphi = r^{\lambda+1} F(\theta)$. $\nabla^4\varphi = 0 \Rightarrow F = A\cos(\lambda+1)\theta + B\cos(\lambda-1)\theta + C\sin(\lambda+1)\theta + D\sin(\lambda-1)\theta$.
2. Symmetry (Mode I) keeps cosine terms; flank conditions $\sigma_{\theta\theta} = \tau_{r\theta} = 0$ at $\theta = \pm\pi$ give two homogeneous equations in A, B.
3. Non-trivial solution needs determinant zero $\Rightarrow$ **$\sin(2\pi\lambda) = 0 \Rightarrow \lambda = n/2$** ($n = 1, 2, 3 \ldots$). Finite energy $\Rightarrow \lambda > 0 \Rightarrow$ lowest is $\lambda = 1/2$ (singular term).
4. Each λ gives one coefficient A_n fixed by the **global** geometry and loading; only A_1 (K) and A_2 (T) have a standard meaning.

B. Plastic-zone size table (plane stress / strain, Irwin)

K (MPa√m)	σ_{ys} (MPa)	r_y plane stress (mm)	r_y plane strain (mm)
30	400	0.90	0.30
50	400	2.49	0.83
80	400	6.37	2.12
50	800	0.62	0.21
100	800	2.49	0.83

- Same $K/\sigma_{ys} \Rightarrow$ same r_y: high-strength alloys have small zones $\Rightarrow$ LEFM works for them; low-strength steels need EPFM.
- **Validity check of LEFM:** require $r_y \ll a, (W-a), B$. Rule of thumb $a, B, (W-a) \geq 2.5(K/\sigma_{ys})^2$ ($\approx 16\ r_y$ in plane stress). For K = 80, σ_{ys} = 400: $(K/\sigma_{ys})^2 = 40$ mm² $\Rightarrow$ need $\geq$ 100 mm sections.

C. Superposition and weight functions

$$K_I = (1/\sqrt{(\pi a)}) \int_{-a}^{a} \sigma(x) \sqrt{((a + x)/(a - x))}\, dx$$

right tip, crack 2a

- $\sigma(x)$ = stress that would exist on the crack line **in the uncracked body**. Handles residual stress, thermal gradients, bending and pressure.

Example (edge crack, small a): membrane 80 MPa + bending (surface) 60 MPa, a = 4 mm: $K = (80 + 60)\times1.12\times\sqrt{(\pi\times0.004)} =$ **17.6 MPa√m**.

Example (weld residual stress): applied 150 MPa and tensile residual stress 200 MPa on an edge crack a = 5 mm: $K_{app} = 21.1$, $K_{res} = 28.1 \Rightarrow K_{total} =$ **49.1 MPa√m** – residual stress contributes more than the applied load! (relieve by PWHT or peening).

D. Choice of specimen and constraint

- Thick, deeply cracked bend/CT specimens: high constraint ($T \geq 0$) ⇒ conservative **K_{Ic}**.
- Thin plates, shallow/ surface cracks, centre-cracked panels: low constraint ⇒ higher apparent toughness; using K_{Ic} is conservative; **constraint-corrected toughness** (J–Q, K–T) can reduce over-conservatism in structural integrity assessment.

E. Dugdale CTOD numerical

$a = 20$ mm, $\sigma/\sigma_{ys} = 0.3$, $\sigma_{ys} = 400$ MPa, $E = 210$ GPa: $\delta = (8\times400\times10^{6}\times0.02/(\pi\times210\times10^{9}))$ $\ln\sec(0.15\pi) = 9.70\times10^{-5} \times 0.1154 =$ **0.0112 mm** (Irwin small-scale: $(4/\pi)K^2/E\sigma_{ys} \approx 0.0137$ mm).

F. Short questions

Q. What does a negative T-stress do? Reduces crack-tip triaxiality (loss of constraint) ⇒ larger plastic zone, higher fracture toughness, stable crack path.

Q. Why does the plastic zone make a crack effectively longer? Yielded material cannot carry the elastic stress; the stress field is displaced forward by r_y, equivalent to a crack of length $a + r_y$.

Q. When is Dugdale preferred to Irwin? Thin sheets in plane stress (steel sheets) with high $\sigma/\sigma_{ys} > 0.5$, where the zone is a narrow strip.

Q. State two uses of CTOD. Fracture criterion for ductile steels (welds, BS 7448), measure of toughness when LEFM is invalid.

Q. Which mode has K_c lowest in most metals? Mode I (plane strain): $K_{Ic} < K_{IIc} < K_{IIIc}$.

G. Practice problems (with answers)

1. Centre crack $2a = 30$ mm, $W = 90$ mm, $\sigma = 120$ MPa. Find K_I. Ans: $\pi a/W = 0.524$, sec = 1.155, $Y = 1.075$ ⇒ $K = 1.075\times120\times\sqrt{(\pi\times0.015)}$ = **28.0 MPa√m**.
2. Find r_y (plane stress & strain) for $K = 60$ MPa√m, $\sigma_{ys} = 500$ MPa. Ans: $(60/500)^2/(2\pi)$ = **2.29 mm**; plane strain **0.76 mm**.
3. Inclined crack $\beta = 45°$, $\sigma = 100$ MPa, $a = 10$ mm. Ans: $K_I = K_{II} = 0.5\times17.72$ = **8.86 MPa√m**; $G = 2\times8.86^2/E'$.
4. Penny-shaped crack radius 3 mm in $\sigma = 200$ MPa. Ans: $K = 2\times200\times\sqrt{(0.003/\pi)}$ = **12.4 MPa√m**.
5. Using Dugdale, find ρ/a for $\sigma/\sigma_{ys} = 0.3$. Ans: $\sec(27°) - 1$ = **0.122**.

4B. Mixed-mode Details

A. General biaxial loading on an inclined crack

$$K_I = (\sigma_y \cos^2\beta + \sigma_x \sin^2\beta)\sqrt{(\pi a)}$$

$$K_{II} = (\sigma_y - \sigma_x)\sin\beta\cos\beta\sqrt{(\pi a)}$$

- β measured from the plane $\perp$ to σ_y. For **equi-biaxial** loading ($\sigma_x = \sigma_y$) $K_{II} = 0$ and $K_I = \sigma\sqrt{(\pi a)}$ for any crack orientation – no mixed mode.

Example: $\sigma_y = 100$, $\sigma_x = 50$ MPa, $\beta = 30°$, $a = 10$ mm: $K_I = (75 + 12.5)\times0.1772 =$ **15.5 MPa√m** ; $K_{II} = 50\times0.433\times0.1772 =$ **3.84 MPa√m**.

B. Where does mixed mode occur?

Situation	Reason
Inclined / misaligned weld defects	Crack not $\perp$ to principal stress
Shafts: torsion + bending	Helical (45°) cracks, Mode I + III
Rolling contact, gears, rails	Shear & compression: Mode II / I + II
Delamination of composites, adhesive joints	Interface cracks always mixed mode
Pressure vessels with nozzles, riveted lap joints	Bending + membrane + shear

C. Strain-energy-density factor (Sih) – preview

$$S = a_{11} K_I^2 + 2\, a_{12} K_I K_{II} + a_{22} K_{II}^2 + a_{33} K_{III}^2$$

- $a_{11} = (1/16\pi\mu)(1+\cos\theta)(\kappa - \cos\theta)$; $a_{12} = (1/16\pi\mu)\sin\theta\,[2\cos\theta - (\kappa-1)]$; $a_{22} = (1/16\pi\mu)[(\kappa+1)(1-\cos\theta) + (1+\cos\theta)(3\cos\theta - 1)]$; $a_{33} = 1/(4\pi\mu)$.
- Strain energy density near tip $dW/dV = S/r$. **Crack grows toward the minimum of $S(\theta)$**; fracture when $S_{min} = S_c$. For pure Mode I S_{min} is at $\theta = 0$.

D. Summary of near-tip fields on the crack line ($\theta = 0$)

	Mode I	Mode II	Mode III
Non-zero stress	$\sigma_x = \sigma_y = K_I/\sqrt{(2\pi r)}$	$\tau_{xy} = K_{II}/\sqrt{(2\pi r)}$	$\tau_{yz} = K_{III}/\sqrt{(2\pi r)}$
K for centre crack	$\sigma\sqrt{(\pi a)}$	$\tau\sqrt{(\pi a)}$	$\tau\sqrt{(\pi a)}$
Energy release G	K_I^2/E'	K_{II}^2/E'	$K_{III}^2/2\mu$
Opening	normal (v)	sliding (u)	tearing (w)

E. Crack-tip blunting – physical picture of CTOD

- A real crack tip blunts under load: tip radius grows from $\rho \approx 0$ to $\rho \approx \delta/2$. The opening at the original tip position δ_t is a measure of **local strain** at the tip.
- In SSY, CTOD ↔ K ↔ J ↔ G are interchangeable: $\delta_t = K^2/(m\,\sigma_{ys}\,E') = G/(m\,\sigma_{ys}) = J/(m\,\sigma_{ys})$, with $m \approx 1$ (plane stress) to 2 (plane strain) (Dugdale $m = 1$).
- CTOD is the fracture parameter of choice for **structural steels / welds** (BS 7448, ISO 12135): fracture when $\delta = \delta_c$.

4C. Irwin's Derivation in Detail & Typical Toughness Values

A. Equilibrium argument for the zone size $2r_y$

1. Elastic stress on the crack line: $\sigma_y = K/\sqrt{(2\pi r)}$. It equals σ_{ys} at $r = r_y = K^2/(2\pi\sigma_{ys}^2)$.
2. Truncating σ_y at σ_{ys} removes load: area under the elastic curve for $0 < r < r_y$ is $\int K/\sqrt{(2\pi r)}dr = 2K\sqrt{(r_y/2\pi)} = 2\sigma_{ys}\,r_y$, while the truncated stress carries only $\sigma_{ys}\,r_y$.
3. The missing force $\sigma_{ys}\,r_y$ must be carried ahead. To keep equilibrium the zone extends to R with $\sigma_{ys}\,R = 2\sigma_{ys}\,r_y \Rightarrow$ **$R = 2r_y$**.
4. The stress curve outside the zone is the elastic one shifted by r_y: effective tip at $a + r_y$, so $K_{eff} = Y\sigma\sqrt{(\pi(a + r_y))}$.
5. Plane strain: constraint raises σ_y at yield to $\approx \sqrt{3}\sigma_{ys}$ (Tresca/von Mises, $\sigma_z = \nu(\sigma_x + \sigma_y)$) ⇒ r_y reduces by 3.

B. Typical fracture toughness K_{Ic} (room temperature)

Material	σ_{ys} (MPa)	K_{Ic} (MPa√m)
Aluminium 7075-T6	500	≈ 29
Aluminium 2024-T3	350	≈ 34
Titanium Ti-6Al-4V	900	≈ 55 – 90
High-strength steel 4340 (tempered low)	1500	≈ 50
Mild / structural steel (above DBTT)	250	> 100 (EPFM required)
Alumina, Si_{3N4} ceramics	—	≈ 3 – 6
Glass / PMMA	—	≈ 0.7 / 1.0 – 1.5
Concrete	—	≈ 0.2 – 1.4

- Trade-off: higher σ_{ys} → lower K_{Ic}. Critical size for 7075-T6 at σ = 300 MPa ($Y = 1$): $a_c = (1/\pi)(29/300)^2 =$ **3.0 mm** — very small ⇒ NDT must detect millimetre cracks; mild steel tolerates much larger flaws.
- Plane-zone check for 7075-T6 at fracture: $r_y(\text{strain}) = (1/6\pi)(29/500)^2 =$ **0.18 mm**,

$<< B$ and $a \Rightarrow$ LEFM valid.

5. Exam Pointers (Unit IV)

1. Explain the Williams series and the T-stress. How does T affect constraint and fracture toughness?
2. Write the near-tip stress field for Modes II and III. Derive K_I, K_{II} for an inclined crack.
3. Evaluate K for SENT / CT / penny-shaped / surface crack (numerical).
4. Derive Irwin's plastic-zone size (plane stress & strain) and the effective crack length. Compute K_{eff}.
5. Derive Dugdale's plastic strip length and CTOD; compare with Irwin.

Formula sheet – Unit IV

> $K = Y\sigma\sqrt{(\pi a)}$; CCT $Y = \sqrt{\sec(\pi a/W)}$; edge $Y = 1.12$; penny $K = 2\sigma\sqrt{(a/\pi)}$

> $r_y = (1/2\pi)(K/\sigma_{ys})^2$ (σ) ; $(1/6\pi)(K/\sigma_{ys})^2$ (ε) ; $a_{eff} = a + r_y$

> Dugdale $\rho = a[\sec(\pi\sigma/2\sigma_{ys}) - 1]$; $\delta = (8\sigma_{ys}\, a/\pi E) \ln \sec(\pi\sigma/2\sigma_{ys}) \rightarrow K^2/E\sigma_{ys}$

> Inclined crack: $K_I = \sigma\sqrt{(\pi a)}\cos^2\beta$; $K_{II} = \sigma\sqrt{(\pi a)}\, \sin\beta\, \cos\beta$

UNIT 5

Toughness Testing, Fatigue Crack Growth, J-Integral, FAD, Mixed Mode, Arrest & Repair

K_{Ic} test | Paris & sigmoidal curve | closure | models | J | FAD | mixed mode | repair

1. Fracture Toughness Testing

Standard	Measures	Specimen
ASTM E399	Plane-strain K_{Ic} (LEFM)	CT, SENB (also arc, disc)
ASTM E561	K–R curve (thin sheet)	M(T), CT
ASTM E1820	J_{Ic}, J–R, CTOD δ (EPFM)	CT, SENB
BS 7448 / ISO 12135	CTOD δ_c, J, K_{Ic}	SENB, CT
ASTM E23	Charpy impact energy, DBTT	10×10 mm V-notch bar
ASTM E647	Fatigue crack growth da/dN–ΔK	CT, M(T)

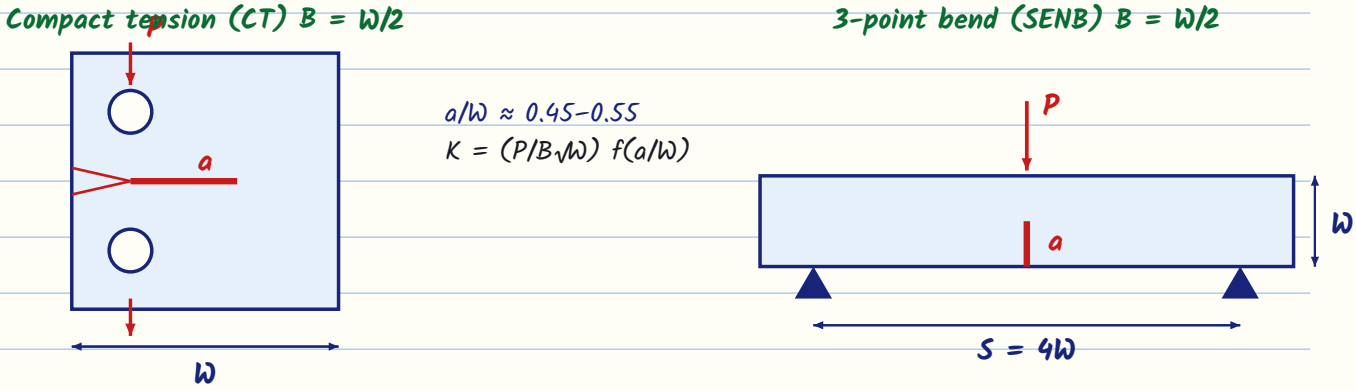
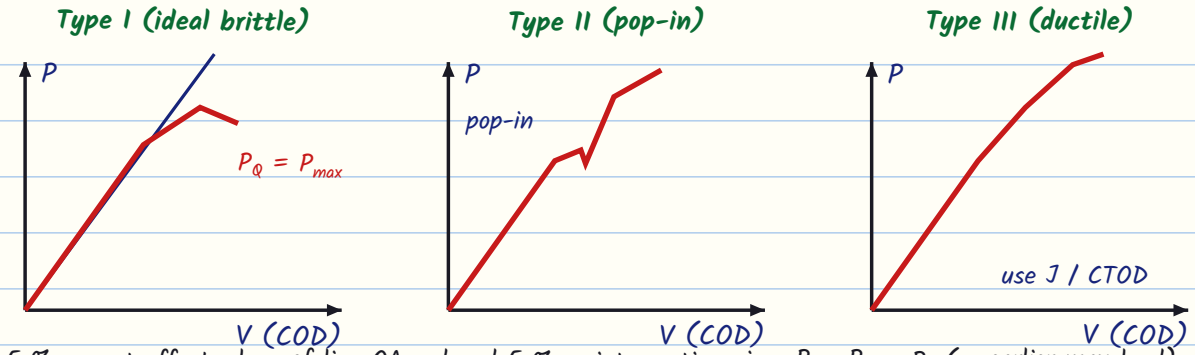


Fig 5.1 Standard K_{Ic} specimens

1.1 E399 procedure

1. Machine specimen: W = 2B, a/W ≈ 0.5 (0.45–0.55), notch with chevron / slot, orientation noted (L-T, T-L).
2. **Fatigue pre-crack** to sharp crack: final $K_{max} \le 0.6\ K_Q$ (to avoid plasticity), ΔK small, length ≥ 1.3 mm.
3. Load at constant displacement rate (K rate 0.55–2.75 MPa√m/s); record P vs crack-mouth opening displacement V with clip gauge.
4. Draw 5 % secant line (slope 0.95 of initial slope); P_5 = intersection; **P_Q** by record type (Fig 5.2); P_{max} = max load.
5. $K_Q = (P_Q/B\sqrt{W})\ f(a/W)$; $K_Q = K_{Ic}$ only if validity criteria are met.



5 % secant offset: slope of line OA reduced 5 % → intersection gives P_5 ; $P_Q = P_5$ (or earlier max load)

Fig 5.2 Load–COD records and P_Q

Validity requirements ($K_Q \rightarrow K_{Ic}$)

> $P_{max}/P_Q \leq 1.10$ (limited non-linearity).

> **$B, a \geq 2.5 (K_Q/\sigma_{ys})^2$** (plane strain; ensures small plastic zone, 50 × r_y).

> $0.45 \leq a/W \leq 0.55$; crack front straightness (a at surface and centre within 10 %).

> If invalid → report K_Q only; use thicker specimen or EPFM (J, CTOD).

Worked Example 1 (E399 check)

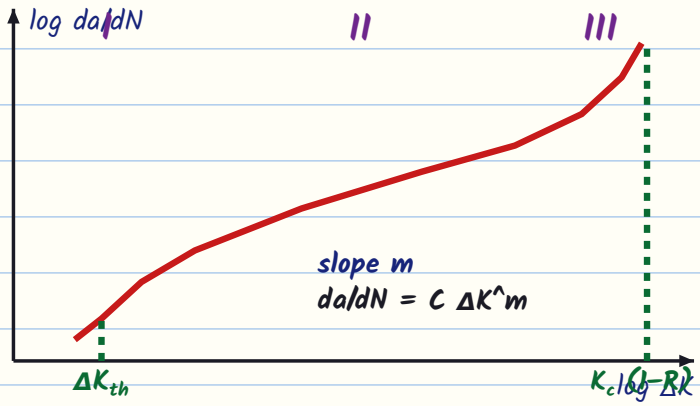
CT: W = 50 mm, B = 25 mm, a = 25.5 mm (α = 0.51), P_Q = 15.8 kN, P_{max} = 16.6 kN, σ_{ys} = 450 MPa. f(0.51) = 9.96.

$K_Q = 15800 \times 9.96/(0.025 \times \sqrt{0.05})$ = **28.2 MPa√m**. Requirement $2.5(28.2/450)^2$ = **9.8 mm**; B = 25 mm and a = 25.5 mm exceed it ✔. P_{max}/P_Q = 1.05 ≤ 1.10 ✔ ⇒ **K_{Ic} = 28.2 MPa√m** (valid).

1.2 J_{Ic} and CTOD testing

- **J_{Ic} (E1820):** single-specimen unloading-compliance (or multiple specimens); plot J vs Δa; blunting line $J = 2\sigma_Y \Delta a$; 0.2 mm offset line ⇒ J_Q; valid if B, $b_0 > 10\ J_Q/\sigma_Y$; $K_{Jc} = \sqrt{(E' J_{Ic})}$.
- **CTOD (BS 7448):** SENB with clip gauge; plastic hinge model: $\delta = K^2(1-\nu^2)/(2\sigma_{ysE}) + r_p(W-a)V_p/[r_p(W-a) + a + z]$, $r_p \approx 0.4$ (rotation factor), V_p = plastic part of V, z = knife-edge height. Reported δ_c (brittle), δ_u, δ_m.
- **Charpy** gives energy-temperature curve (DBTT), has empirical correlations with K_{Ic} (Rolfe–Barsom) – used only for screening, not for design values.

2. Paris Law and the Sigmoidal Curve



Region I : threshold, microstructure & R-sensitive, $da/dN < 10^{-9}$ m/c

Region II : Paris law (log-linear) stable, $m = 2 - 4$

Region III: $K_{max} \to K_c$, rapid, static modes, thickness-sensitive

Fig 5.3 Sigmoidal $da/dN - \Delta K$ curve (log–log)

Region	Behaviour	Controlled by
I (near threshold)	da/dN falls steeply to ~10^{-10} m/cycle at ΔK_{th}; below it no growth	microstructure, mean stress R, environment, closure
II (intermediate)	log da/dN vs log ΔK straight line: $da/dN = C \Delta K^m$	ΔK (stable, most of the life)
III (high ΔK)	da/dN accelerates as $K_{max} \to K_c$ (R) ; static fracture modes	K_{max}, K_c, thickness, microstructure

- **Paris–Erdogan (1963):** $da/dN = C (\Delta K)^m$; m = slope in log–log plot (2–4 metals, 3 ≈ steel, 3–4 Al; ceramics 10–50). C depends on units, R, environment.
- $\Delta K = K_{max} - K_{min} = \Delta\sigma\, Y \sqrt{(\pi a)}$ (for $R \geq 0$). Threshold $\Delta K_{th} \approx 6.4(1 - 0.85R)$ MPa√m for steels, ≈ 2–4 (Al) — below ΔK_{th}: infinite life (fatigue-limit analogue for cracks).

Material class (Barsom–Rolfe)	C (m/cycle, MPa√m)	m
Ferrite–pearlite steels	6.9×10^{-12}	3.0
Martensitic steels	1.35×10^{-10}	2.25
Austenitic stainless	5.6×10^{-12}	3.25
Al alloys (typical)	$\approx 5 \times 10^{-11}$	≈ 3–3.5

Worked Example 2 (Paris life)

Edge crack, $Y = 1.12$, $\Delta\sigma = \sigma_{max} = 120$ MPa ($R = 0$), $a_0 = 1$ mm, $K_c = 100$ MPa√m, ferrite-pearlite steel ($C = 6.9 \times 10^{-12}$, $m = 3$).

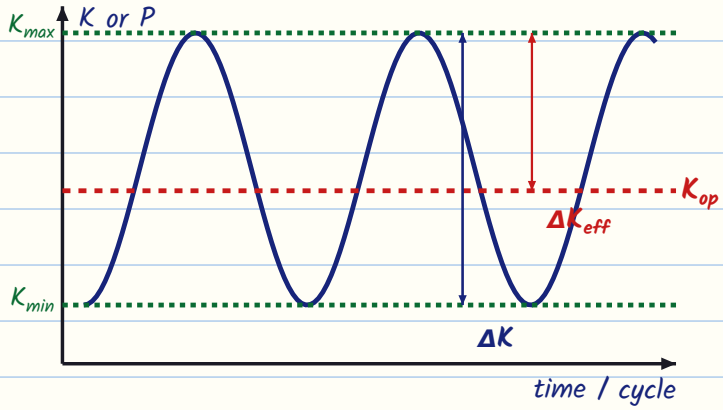
$a_c = (1/\pi)(100/(1.12 \times 120))^2 =$ **0.176 m**. $N_f = (a_c^{-0.5} - a_0^{-0.5})/[C(Y\Delta\sigma\sqrt{\pi})^3(-0.5)] = (2.382 - 31.62)/(6.9 \times 10^{-12} \times (238.2)^3 \times (-0.5)) \approx$ **6.3×10^5 cycles**.

Inspection: $N_f/4 \approx 1.6 \times 10^5$ cycles; $\Delta K_0 = 120 \times 1.12 \times \sqrt{(\pi \times 0.001)} = 7.5$ MPa√m > ΔK_{th} (5.9) ⇒ crack will grow.

2.1 Effects of R, environment, thickness

- ↑R ⇒ ↑da/dN, ↓ΔK_{th} (Region I & III most sensitive; Region II mildly).
- Corrosive environment / hydrogen / high T ⇒ faster, lower ΔK_{th}. High frequency tends to reduce environmental effect.
- Large grains, high yield & rough crack path ⇒ higher ΔK_{th} (roughness closure).

3. Crack Closure



Elber (1970):

crack faces touch (closed) before load reaches zero (wake plasticity)

$\Delta K_{eff} = K_{max} - K_{op}$

$U = \Delta K_{eff}/\Delta K = 0.5 + 0.4R$

$da/dN = C\,(U\,\Delta K)^m$

(−0.1 ≤ R ≤ 0.7, Al 2024-T3)

Fig 5.4 Elber's crack closure concept

- **Elber (1970):** fatigue cracks remain closed during part of the tensile loading because of residual plastic stretch left in the wake of the crack. Only the portion above K_{op} is effective.

$$\Delta K_{eff} = K_{max} - K_{op}\ ,\ U = \Delta K_{eff}/\Delta K = 0.5 + 0.4R\ (-0.1 \le R \le 0.7)$$

$$da/dN = C\,(\Delta K_{eff})^m = C\,(U\,\Delta K)^m$$

Closure mechanism	Cause	Where important
Plasticity-induced	residual stretch in the wake	Region II, plane stress, thin sheet
Oxide-induced	oxide debris/ corrosion product wedges crack	Region I, humid / high T, low R
Roughness-induced	mismatch of fracture surface asperities (Mode II shifts)	Region I, coarse grains, low R
Viscous-fluid / phase transformation	fluid in crack; martensite transformation zone	oils, TRIP steels, ZrO_2

Worked Example 3

$R = 0.3$, $\Delta K = 15$ MPa$\sqrt{m} \Rightarrow U = 0.5 + 0.4 \times 0.3 = 0.62 \Rightarrow \Delta K_{eff} =$ **9.3 MPa$\sqrt{m}$**; growth rate is $(0.62)^m$ times that of $\Delta K = 15$ without closure (for $m = 3 \rightarrow 24$ %).

3.1 Variable amplitude – overloads

- **Single tensile overload** (OLR = $K_{ol}/K_{max} \approx 1.5$–2): large plastic zone ⇒ compressive residual stress & increased closure ⇒ **retardation** (delay), possibly arrest at OLR ≥ 2.
- Retardation lasts until the crack grows through the overload plastic zone r_{ol}. **Underloads** and compressive loads reduce/negate retardation. Spectrum loading (rainflow cycle counting + damage summation) is used in practice.

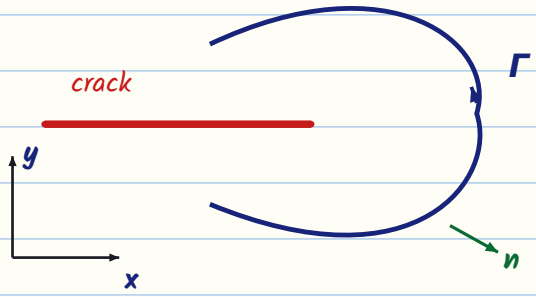
4. Crack Growth Models

Model	Equation	Remarks
Paris–Erdogan	$da/dN = C\,\Delta K^m$	Region II, constant R
Walker (1970)	$da/dN = C[\Delta K/(1-R)^{1-\gamma}]^m = C[K_{max}(1-R)^\gamma]^m$	R-effect; $\gamma \approx 0.5$–0.9 fitted
Forman (1967)	$da/dN = C\,\Delta K^m / [(1-R)K_c - \Delta K]$	includes Region III, R
Elber	$da/dN = C\,(U\,\Delta K)^m$	closure-based
Priddle / modified Forman	$da/dN = C\,\Delta K^m\,(1 - \Delta K_{th}/\Delta K)^p / (1 - K_{max}/K_c)^q$	Regions I-III (sigmoidal)
NASGRO	$da/dN = C[(1-f)/(1-R)\,\Delta K]^n\,(1-\Delta K_{th}/\Delta K)^p / (1-K_{max}/K_c)^q$	f = closure function; used in NASA/ESA codes
Wheeler (1972)	$da/dN = C_p\,(da/dN)_{CA}$; $C_p = [r_{pi}/(a_{ol} + r_{ol} - a_i)]^p$	overload retardation, p fitted
Willenborg	$K_{eff} = K_{max} - K_{comp}$ (residual stress intensity)	retardation, no fit constant (generalised)

4.1 Using the models

- Constant amplitude: Paris if one R; Walker/Forman for several R; NASGRO when sigmoidal behaviour/threshold matters (damage-tolerance design of aircraft).
- Variable amplitude: **equivalent ΔK_{rms}** = $[\Sigma\, n_i\,\Delta K_i^m / \Sigma\, n_i]^{1/m}$ or cycle-by-cycle integration with Wheeler / Willenborg / closure (FASTRAN, AFGROW, NASGRO software).
- **Wheeler example:** $r_{pi} = 0.5$ mm, overload zone $r_{ol} = 2$ mm, $a_{ol} = 10$ mm, $a_i = 10.2$ mm, $p = 1.5$: $C_p = [0.5/(10 + 2 - 10.2)]^{1.5} = (0.278)^{1.5} =$ **0.146** ⇒ growth rate reduced to 15 %.
- Life = inspection interval: with probability of detection (POD) curve & scatter factor; Retirement for cause.

5. J-Integral



Rice (1968):

$$J = \oint_\Gamma \left(w\, dy - T_i \frac{\partial u_i}{\partial x} ds \right)$$

$w = \int \sigma_{ij}\, d\varepsilon_{ij}$ (strain energy density)
$T_i = \sigma_{ij}\, n_j$ (traction on Γ)

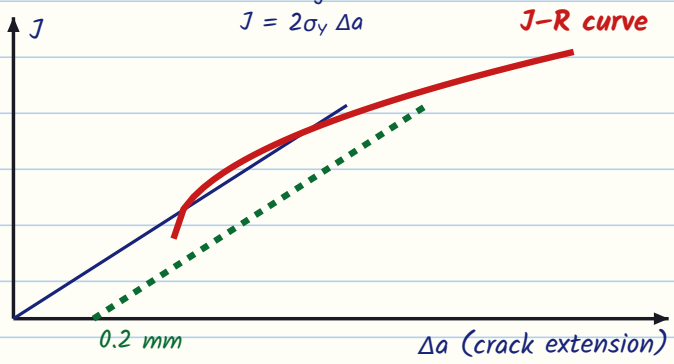
- path independent (non-linear elastic)
- $J = -d\Pi/da = G$ (linear elastic)
- $J = 0$ on closed path w/o singularity

Fig 5.5 Contour for the J-integral

- For a **non-linear elastic** material (deformation plasticity), J is path independent. Physical meaning: **$J = -(1/B)\, d\Pi/da$** — energy release rate for non-linear elastic bodies. Linear elastic: **$J = G = K^2/E'$**.
- HRR field (Hutchinson, Rice, Rosengren) for Ramberg–Osgood material $\varepsilon/\varepsilon_0 = \sigma/\sigma_0 + \alpha(\sigma/\sigma_0)^n$:

$$\sigma_{ij} = \sigma_0 \left[J / (\alpha\, \varepsilon_0\, \sigma_0\, I_n\, r) \right]^{1/(n+1)} \tilde{\sigma}_{ij}(\theta, n)$$

- J **uniquely** characterises the stress/strain singularity ($\sigma \sim r^{-1/(n+1)}$, $\varepsilon \sim r^{-n/(n+1)}$); $n = 1$ recovers LEFM ($r^{-1/2}$). Fracture at $J = J_{Ic}$.
- **Relations:** $\delta_t = J/(m_1 \sigma_{ys})$ ($m_1 \approx 1$–2.5); $K_{Jc} = \sqrt{(E' J_c)}$; $J = J_{el} + J_{pl}$; $J_{el} = K^2/E'$.



J_{Ic} (ASTM E1820):

intersection J_Q of R-curve with 0.2 mm offset line (parallel to blunting line)

valid if $B, b_0 > 10\, J_Q/\sigma_Y$

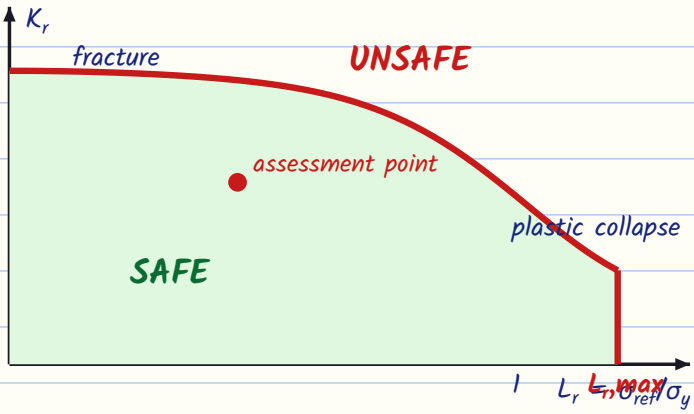
Tearing modulus $T = (E/\sigma_0^2)\, dJ/da$
Stable growth: $T_{app} < T_{mat}$

Fig 5.6 J–R curve and determination of J_{Ic}

- **EPRI estimation:** $J = J_e(a_{eff}) + \alpha\, \varepsilon_0\, \sigma_0\, b\, (a/W)\, h_1(a/W, n)\, (P/P_0)^{n+1}$ (b = ligament, P_0 = limit load).
- Limitations: no unloading (deformation theory), small crack growth $\Delta a \ll b \Rightarrow$ J-controlled growth $\omega = (b/J)(dJ/da) \gg 1$.

Example: $K = 60$ MPa$\sqrt{m}$ (plane strain, $E = 210$ GPa, $\nu = 0.3$): $J = (1-\nu^2)K^2/E = 0.91 \times (60\times10^6)^2/(210\times10^9) =$ **15.6 kJ/m²**; $\delta = J/(2\sigma_{ys}) = 15600/(2\times400\times10^6) =$ **0.0195 mm**.

6. Failure Assessment Diagram (FAD)



Option 1 (BS 7910 / R6):

$K_r = K_I/K_{mat}$ (+ρ)

$L_r = \sigma_{ref}/\sigma_y = P/P_L$

$f(L_r) = (1-0.14L_r^2)$ × $[0.3 + 0.7\ \exp(-0.65\ L_r^6)]$

cut-off $L_{r,max} = (\sigma_y+\sigma_u)/2\sigma_y$

Fig 5.7 FAD – two-criteria approach (R6 / BS 7910 Option 1)

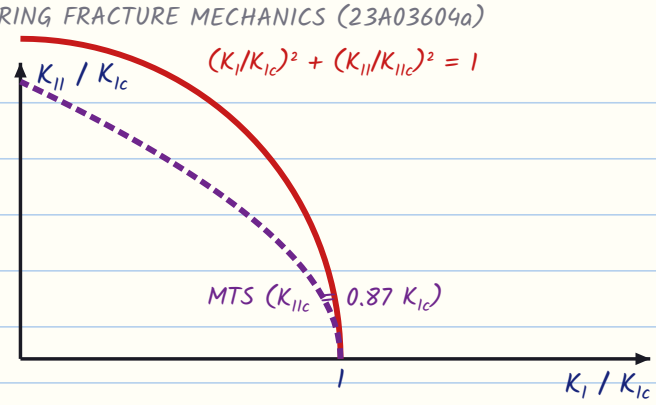
- Interpolates between **brittle fracture (K_r = 1)** and **plastic collapse (L_r = 1)**. Components with flaws are assessed by a single point (L_r, K_r).

1. Characterise flaw (a, shape) from NDT; apply conservative flaw-size interaction/ recategorisation rules.
2. Calculate K_I (primary + secondary stresses, with plasticity correction ρ) ⇒ **$K_r = K_I/K_{mat}$** ($K_{mat} = K_{Ic}$, or from J/δ).
3. Reference stress σ_{ref} (net-section) ⇒ **$L_r = \sigma_{ref}/\sigma_y = P/P_L$**.
4. Plot (L_r, K_r). Inside the failure line $K_r = f(L_r)$ and $L_r < L_{r,max}$ ⇒ **acceptable**; on/outside ⇒ unacceptable (repair / re-rate / reduce load).

Example: K_I = 50, K_{mat} = 100 MPa√m ⇒ K_r = 0.50; σ_{ref} = 240, σ_y = 400 ⇒ L_r = 0.60. $f(0.6) = 0.9496 \times 0.979 = 0.93 > 0.50$ ⇒ **safe**. Reserve factor along the ray ≈ 1.46 (load can rise ~46 % before the assessment line is reached).

- Cut-off $L_{r,max} = (\sigma_y + \sigma_u)/(2\sigma_y)$ (e.g. 1.2 for σ_u/σ_y = 1.4). Option 2 & 3 use actual stress–strain curve / J-based FAD (more accurate, less conservative).
- Used in BS 7910, API 579, R6, SINTAP, FITNET: fitness-for-service of pressure vessels, pipelines, nuclear and offshore structures.

7. Mixed-Mode Fracture



Mixed-mode envelopes (I + II)

Inside the curve : no fracture
Outside / on curve : fracture

Kink angle (MTS):

$\theta_0 = 2\tan^{-1}[(K_I - \sqrt{(K_I^2+8K_{II}^2)})/4K_{II}]$

pure II : $\theta_0 = -70.5°$

Fig 5.8 Mixed-mode I + II failure envelopes

Criterion	Statement	Result for pure Mode II
Max tangential stress (MTS, Erdogan-Sih)	Crack grows ⊥ to max $\sigma_{\theta\theta}$; at θ_0 : $K_I \sin\theta_0 + K_{II}(3\cos\theta_0 - 1) = 0$; fracture when $\cos(\theta_0/2)[K_I \cos^2(\theta_0/2) - 1.5K_{II}\sin\theta_0] = K_{Ic}$	$\theta_0 = -70.5°$, $K_{IIc} = 0.87 K_{Ic}$
Strain-energy density (SED, Sih)	Growth toward S_{min} ; fracture when $S_{min} = S_c$	$K_{IIc} \approx 1.05\ K_{Ic}$ ($\nu = 0.3$)
Max energy release rate (Hussain)	Growth in direction of G_{max} for kinked crack	$\theta_0 \approx -75°$, $K_{IIc} = K_{Ic}$
Simple interaction	$(K_I/K_{Ic})^2 + (K_{II}/K_{IIc})^2 = 1$; or $G_I + G_{II} = G_c$	—

- Equivalent K: $K_{eq} = \sqrt{(K_I^2 + K_{II}^2)}$ (conservative, from $G = G_I + G_{II}$), or MTS-based K_{eq} above. Mode III adds $K_{III}^2(1+\nu)$.

Example (Unit IV, Ex.1): $K_I = 13.29$, $K_{II} = 7.67 \Rightarrow \theta_0 = -43.2°$; $K_{eq}(MTS) = \cos(\theta_0/2)[K_I \cos^2(\theta_0/2) - 1.5K_{II}\sin\theta_0] \approx$ **18.0 MPa√m** (vs $\sqrt{(K_I^2+K_{II}^2)} = 15.3$). Fracture if $K_{eq} \geq K_{Ic}$.

- Experiments: inclined centre-cracked plate, CTS (Richard) specimen, Arcan fixture, 4-point-bend SENB; fatigue growth of mixed-mode cracks: ΔK_{eq}-Paris, crack kinks and then grows in near Mode I.

8. Crack Arrest and Repair

8.1 Crack arrest

- A running (dynamic) crack stops when the dynamic driving force K_I falls below the **arrest toughness K_{Ia}** ($< K_{Ic}$). Approaches: tough steel (low DBTT), lower stress regions, **crack arrester** strips/ stiffener, riveted seams (Liberty-ship lessons), tough inserts in pipelines, crack arrest rings.
- Tests: Robertson test (crack arrest temperature CAT), ESSO test, Double-cantilever (K_{Ia}, ASTM E1221), drop-weight NDT temperature (E208). Fail-safe: rivet/ stringer-frame arrangement limits K as crack approaches a stiffener (K falls).

8.2 Repair methodologies

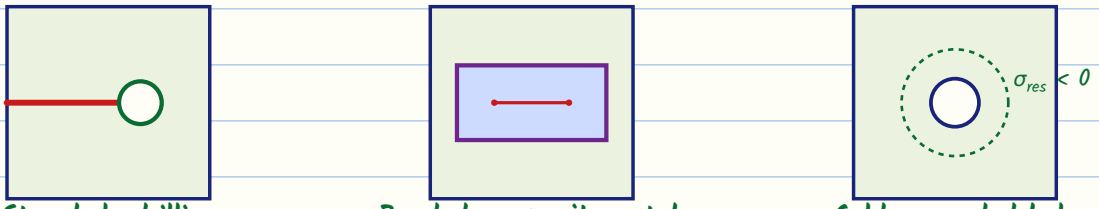


Fig 5.9 Typical repair techniques

Method	Principle / remarks
Stop-hole drilling	Drill hole (radius ρ) just ahead of tip: blunts crack, $K_t \approx 1 + 2\sqrt{(a/\rho)}$. Temporary; cold-work or insert fastener to extend life.
Grind-out & re-weld	Remove crack fully (verify by DPT/MPI/UT), V-groove, weld with low-hydrogen electrodes, PWHT / peen.
Bonded composite patch (boron/epoxy, CFRP)	Bridges crack, reduces ΔK 50-90 %, little added stress; used on aircraft (F-111, RAAF), pipelines.
Cold working / mandrel expansion	Residual compressive stress around hole (2–6 % expansion) ⇒ fatigue life ×3–10.
Shot / laser / hammer peening, HFMI	Surface compressive stress; delays initiation & growth at weld toes.
Reinforcement: doublers, sleeves, clamps	Reduce local stress; pipeline composite wrap, Type B sleeve.
Stress relief / PWHT	Reduces tensile residual stress (K_{res}).

- **Inspection** (NDT): dye penetrant, magnetic particle, eddy current, ultrasonic (TOFD, phased array), radiography. Damage-tolerance plan: $a_{detectable} \rightarrow a_c$; interval ≤ N/(scatter factor 2–4).
- **Repair decision:** assess by FAD / Paris life; if not acceptable → repair, derate load, or retire.

9A. Additional Notes & Examples

A. Fatigue crack growth testing (ASTM E647)

1. CT or M(T) specimen, fatigue pre-crack to a_0 at the test ΔK; constant-amplitude load, R fixed (0.1 typical), f = 10–30 Hz.
2. Monitor a: travelling microscope, compliance (clip gauge), potential drop (DCPD) – record (a, N) pairs.
3. da/dN by **secant** ($\Delta a/\Delta N$ between points) or **7-point incremental polynomial**; ΔK from handbook $Y(a/W)$.
4. Near threshold: **load-shedding** (ΔK-decreasing, $C = (1/\Delta K)(d\Delta K/da) \approx -0.08\ mm^{-1}$) until da/dN ~ 10^{-10} m/cycle $\Rightarrow \Delta K_{th}$.
5. Plot log da/dN vs log ΔK $\Rightarrow$ fit C, m (Region II), ΔK_{th}, K_c.

B. Numerical – effect of R (Walker vs Paris)

Walker with $C = 3\times10^{-12}$, m = 3, γ = 0.5: for ΔK = 20 MPa$\sqrt{m}$ at R = 0.5, $\Delta K/(1-R)^{0.5}$ = 28.3 $\Rightarrow$ da/dN = $3\times10^{-12} \times 28.3^3$ = **6.8×10^{-8} m/cycle**; at R = 0 : $3\times10^{-12}\times20^3 = 2.4\times10^{-8}$.

Ratio 2.83 $\Rightarrow$ the mean-stress effect nearly triples the growth rate.

Forman (illustrative): $C = 3\times10^{-10}$, m = 3, K_c = 60, R = 0.5, ΔK = 20 $\Rightarrow$ da/dN = $3\times10^{-10}\times8000/(0.5\times60 - 20)$ = **2.4×10^{-7}**; as $\Delta K \to (1-R)K_c$ = 30 the denominator $\to$ 0 and da/dN $\to \infty$ (Region III).

C. Tearing modulus (ductile stable growth)

- $T_{mat} = (E/\sigma_0^2)(dJ/da)$. Stable growth requires $T_{app} < T_{mat}$; instability when the applied tearing modulus $T_{app} = (E/\sigma_0^2)(\partial J/\partial a)_\Delta$ equals T_{mat}.

Example: E = 210 GPa, σ_0 = 400 MPa, dJ/da = 200 MN/m² $\Rightarrow$ $T = (210\times10^9/(400\times10^6)^2) \times 2\times10^8$ = **263** (typical for tough structural steels: 50–500).

D. FAD – further points

- **Secondary stresses** (residual, thermal) raise K but do not contribute to plastic collapse: $K_r = (K_p + K_s)/K_{mat} + \rho$, where ρ is a plasticity interaction term (small at low L_r).
- **Reference stress** examples: $\sigma_{ref} = P/(B(W-a))$ × factor for CT; for a through crack in a plate $\sigma_{ref} = \sigma_{net}$-section ($\sigma$ W/(W − 2a)).
- A flaw that is unacceptable may be accepted after more refined analysis (Option 2/3), better K_{mat} (CTOD/J data) or removing conservatism (actual σ_y).

E. Two-mark answers

Q. Why is fatigue pre-cracking required? To obtain a natural, sharp crack (root radius $\to$ 0) so measured toughness is a lower-bound material property.

Q. What is ΔK_{th}? Stress-intensity range below which fatigue cracks do not propagate (da/dN < 10^{-10} m/cycle).

Q. State two limitations of Paris law. Valid only for Region II; ignores R, threshold, Region III and load-sequence effects.

Q. What is J in linear elasticity? $J = G = K^2/E'$.

Q. What does a point outside the FAD curve mean? The flaw is unacceptable: failure by fracture/collapse is predicted.

Q. Name two crack-repair methods that induce compressive stress. Cold-expansion of holes and peening.

Q. What is retardation? Temporary reduction of crack growth rate after a tensile overload.

9B. Damage-Tolerance Design Example

A. Residual-strength curve

Edge crack, $Y = 1.12$, $K_c = 100$ MPa$\sqrt{m}$: $\sigma_f = K_c/(Y\sqrt{(\pi a)})$. (Capped by yield/net-section collapse, e.g. 400 MPa.)

a (mm)	Failure stress σ_f (MPa)
5	712
10	504
20	356
50	225
100	159

- Strength falls rapidly as the crack grows; at $a = 50$ mm the part fails at only 225 MPa. Small cracks ($a \leq 10$ mm) are limited by plastic collapse rather than fracture.

B. Inspection interval

Same steel as Example 2 ($C = 6.9\times10^{-12}$, $m = 3$, $\Delta\sigma = 120$ MPa, $Y = 1.12$). NDT can reliably detect $a_d = 2$ mm. Life from 2 mm to $a_c = 176$ mm: $N = (0.176^{-0.5} - 0.002^{-0.5})/(C(238.2)^3(-0.5))$ = **4.3e+05 cycles**.

With scatter factor 2 ⇒ inspect every ≈ **2.1e+05 cycles** (e.g. ~ 2 years at 10^5 cycles/year). If a crack ≥ a_d is found ⇒ repair (stop-hole/ weld/ patch) or replace.

C. Stop-hole numerical

Crack $a = 20$ mm stopped by a hole of radius $\rho = 3$ mm: $K_t \approx 1 + 2\sqrt{(a/\rho)} = 1 + 2\sqrt{(20/3)}$ = **6.2** (vs ∞ for a sharp crack). Larger hole radius ⇒ lower K_t but more stress concentration due to hole itself; typical hole diameter ≈ 3–6 × plate thickness for thin sheet, drilled so that the crack tip lies within the hole.

D. Comparison summary of fracture parameters

Parameter	Applicability	Criterion	Units
K	LEFM, SSY	$K = K_{Ic}$	MPa√m
G	LEFM	$G = G_c$	J/m² (N/m)
CTOD δ	LEFM → EPFM	$\delta = \delta_c$	mm
J	EPFM (non-linear elastic)	$J = J_{Ic}$, $T_{app} < T_{mat}$	kJ/m²
FAD (K_r, L_r)	Engineering assessment	(L_r, K_r) inside curve	—

9. Exam Questions & Formula Sheet (Unit V)

1. Describe ASTM E399 K_{Ic} test: specimen, pre-cracking, P_Q determination and validity criteria. Solve a numerical.
2. Sketch the sigmoidal da/dN–ΔK curve and explain the three regions. State Paris law and estimate life.
3. Explain crack closure (Elber) and its effect on growth; discuss overload retardation.
4. Write Walker, Forman, NASGRO and Wheeler models and state their use.
5. Define J-integral; show path independence idea; state HRR field, J_{Ic} determination and J–CTOD relation.
6. Explain FAD and procedure for assessing a flaw. Solve a numerical.
7. Explain mixed-mode criteria (MTS, SED, G_{max}). Find the kink angle.
8. Discuss crack arrest and repair methods with examples.

Formula sheet – Unit V

> K_{Ic} valid: $B, a \geq 2.5(K_Q/\sigma_{ys})^2$; $P_{max}/P_Q \leq 1.1$; $K_Q = (P_Q/B\sqrt{W})\, f(a/W)$

> Paris: $da/dN = C\,\Delta K^m$; $N = [a_c^{1-m/2} - a_0^{1-m/2}] / [C(Y\Delta\sigma\sqrt{\pi})^m(1-m/2)]$

> Closure: $\Delta K_{eff} = K_{max} - K_{op}$; $U = 0.5 + 0.4R$; Walker, Forman: see §4

> $J = -d\Pi/da$; $J = K^2/E'$ (LEFM) ; $\delta = J/(m\sigma_{ys})$; HRR $\sigma \sim r^{-1/(n+1)}$

> FAD: $K_r = K_I/K_{mat}$; $L_r = \sigma_{ref}/\sigma_y$; $f = (1-0.14L_r^2)[0.3+0.7e^{-0.65L_r^6}]$

> MTS: $\theta_0 = 2\tan^{-1}[(K_I - \sqrt{(K_I^2+8K_{II}^2)})/4K_{II}]$